

KATHMANDU MODEL SECONDARY SCHOOL
PRE - TERM EXAMINATION 2081

Class: XI 'Science' DAY [SET-B] F.M: 35
Time: 1.5 hrs. Subject: Physics [1011] P.M: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP-A MCQ [7×1=7]

- The solar constant is defined as the energy incident per unit area per second on the earth. The dimensions of solar constant are
a. $[M L T^{-2}]$ b. $[M L T^{-3}]$ c. $[M L T^{-1}]$ d. $[M L T^{-2}]$
- Which of the following combinations of three dimensionally different physical quantities P, Q and R can never be a meaningful quantity?
a. $PQ + R$ b. PQ/R c. $(P+Q)/R$ d. $(PQ-R)/QR$
- Find the odd one out among the following in terms of the type of quantity: Displacement, velocity, acceleration, weight, speed
a. Weight b. displacement c. speed d. velocity
- If change in temperature in Celsius scale is 45°C , the temperature change in kelvin scale will be
a. 318K b. 273K c. 0K d. 45K
- How many electrons make up a charge of $2 \times 10^{-18} \text{ C}$?
a. 1.25×10^{18} b. 2.23×10^{14} c. 1.25×10^4 d. 2.23×10^4
- The soap bubble is given a negative charge, then its radius
a. decreases
b. increases
c. remains unchanged
d. decreases or increases on the basis of magnitude of charge.
- A conducting wire is connected between two conducting spheres of equal size have a charge of -3C and $+1\text{C}$ respectively. Find out the new charge on each sphere.
a. -4C b. $+4\text{C}$
c. -1C d. $+1\text{C}$

GROUP-A [4×5=20]

Short answer question.

- a. State principle of homogeneity. [1]
b. A student writes the formula for speed of longitudinal waves as $v = \sqrt{\frac{P}{\rho}}$, where P the pressure D is density of the medium and ρ is the ratio of two

specific heats of gases which has no dimension. Check the correctness of the formula. [2]

- Taking force, length and time to be fundamental quantities, find the dimensional formula of density. [2]
- a. State triangle law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors. [3]
b. Two vectors have equal magnitudes, and their resultant also has the same magnitude. What is the angle between the two vectors? [2]
- a. What are necessary conditions for a physical quantity to be a vector quantity? [1]
b. Two forces of 30N and 40N are inclined to each other at an angle of 60° . What is their resultant? N [2]
c. Show that work done and kinetic energy have same dimensions. [2]
- a. How can you charge a body negatively by induction? Explain. [3]
b. Explain the meaning of the quantization of charge. [1]
c. Does charging by friction follows conservation of charge? [1]

GROUP-A [1×8=8]

Long answer questions.

- Define temperature on the basis of zeroth law of thermodynamics. [1]
a. If two bodies are in thermal equilibrium in one frame, will they lie in the thermal equilibrium in all frames? Explain. [2]
b. The steam point and ice point of a mercury thermometer are marked as 80° and 20° respectively. What will be the temperature in centigrade scale when this thermometer read 32° ? [3]
c. Write short note about absolute scale of temperature. [2]

"Best of luck"

KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION 2081
(DAY) SET-A

Class: XI 'Science' F.M: 35
Time: 1.30 hrs. Subject: Chemistry [1011] P.M: 14
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' MCQ [5×1=5]

- Which of the following is not a redox reaction?
a) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ b) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
c) $\text{Ni} + \text{H}_2\text{O} \rightarrow \text{NiOH} + \frac{1}{2}\text{H}_2$ d) $\text{MnCl}_2 \rightarrow \text{MnCl}_3 + \frac{1}{2}\text{Cl}_2$
- The oxidation number of Fe in the compound Fe_3O_4 is
a) $+4/3$ b) $+2/3$ c) $+5/3$ d) $+8/3$

The unit of universal gas constant is

- Joule stores mol^{-1}
- $\text{Cal K}^{-1} \text{mol}^{-1}$
- $\text{Cal meter degree}^{-1}$
- Joule/mole

Which of the following expression at constant pressure represents Charles's law?

- $V \propto \frac{1}{T}$
- $V \propto \frac{1}{T^2}$
- $V \propto T$
- $V \propto T^2$

Which is correctly arranged in increasing order?

- $F < O < C < Cl < Br$
- $C < O < F < Cl < Br$
- $Cl < Br < F < C < O$
- $O < F < C < Cl < Br$

GROUP 'B' [5×1=5]

- State and explain Boyle's law. [2]
b. What is the percentage increase in the volume of a gas when its pressure is decreased by 20% at constant temperature? [3]
a) Derive $PV = nRT$. Why do real gases deviate from ideal behavior? [3+1]
b) Calculate the value of R in CGS system. [1]
State modern periodic law. What are the advantages of the modern periodic table? Compare the size of Na and Na^+ with proper justification. [1+3+1]
How do you account for the fact that the 1st ionization energy of N is more than that of oxygen. Show the variation of ionization energy in a group and in a period. [2+1.5+1.5]
Define electronegativity and π -electronation with examples. Calculate the O.N. of underlined elements. [2+1]
a) NH_4NO_3 b) $\text{K}_2[\text{Fe}(\text{CN})_6]$ c) NaBH_4
Define auto-redox reaction with an example. Balance the following redox reaction by oxidation number method and also identify oxidant and reductant. [1+4]
 $\text{Zn} + \text{KNO}_3 + \text{KOH} \rightarrow \text{K}_2\text{ZnO}_2 + \text{NH}_4 + \text{H}_2\text{O}$

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PRE-TERM EXAMINATION 2081

Class: XI 'Science' (DAY) SET-B F.M: 35
Time: 1.30 hrs. Subject: Chemistry [1011] P.M: 14

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Attempt all the questions.

GROUP 'A' MCQ [5×1=5]

- Air at sea level is dense. This is a practical application of
a) Boyle's law b) Charles's law c) Dalton's law d) Avogadro's law
- The lowest hypothetical or imaginary temperature at which gases are supposed to occupy zero volume is called
a) Kelvin temperature b) absolute zero
c) Charles's temperature d) Constant temperature
- The correct value of the gas constant 'R' is close to
a) $0.082 \text{ lit atm}^{-1} \text{ mol}^{-1}$ b) $0.082 \text{ lit atm}^{-1} \text{ mol}^{-1}$
c) $0.082 \text{ lit atm}^{-1} \text{ mol}^{-1}$ d) $0.082 \text{ lit atm}^{-1} \text{ mol}^{-1}$
- The correct order of first ionization potential of Be, B, C, N, O is
a) $\text{Be} < \text{B} < \text{C} < \text{N} < \text{O}$ b) $\text{B} < \text{Be} < \text{C} < \text{O} < \text{N}$
c) $\text{Be} < \text{B} < \text{C} < \text{O} < \text{N}$ d) $\text{Be} < \text{B} < \text{C} < \text{N} < \text{O}$
- Manganese exhibits maximum oxidation state in
a) KMnO_2 b) KMnO_4 c) MnO_2 d) Mn_2O_3

GROUP 'B' [5×1=5]

- Answer The Following Questions.
a) Define Boyle's law. Also plot graphs to illustrate Boyle's law. [2+1]
b) Define ideal gas and real gas. [2]
7. a) A flask of 2 litres capacity can withstand a pressure of 16 atm and is filled with CO_2 at 27°C and 760 mmHg pressure. At what temperature will it explode on heating? [1]
b) State the condition at which real gas deviate from ideal behavior. [1]
8. a) State Modern Periodic law and explain the classification of elements on the basis of differentiating electrons. [1+1]
b) Define electron affinity? What are the factors that affect electron affinity? [2]
9. a) How do atomic radii vary in a group & period? [2]
b) Compare the size of F^- and F with reason. [1]
c) What is the screening effect? [1]
10. a) Define Oxidizing agent and reducing agent giving suitable example. [1]
b) H_2S acts as reducing agent why? [1]
c) Show with examples that all reactions are not redox reaction. [2]
11. a) Calculate the oxidation number of the underlined species. i) Fe_3O_4 ii) Fe_2O_3 [2]
b) Balance the following equation by oxidation number method. [4]
 $\text{FeSO}_4 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
"Best of Luck"

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' MORNING [SET-B] F.M: 35
Time: 1.5 hrs. Subject: Physics [1011] P.M: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP 'A' MCQ [7x1=7]

Multiple choice questions

- The dimensions of Power are
a. $[ML^2T^{-3}]$ b. $[M^{-1}L^2T^{-3}]$
c. $[M^{-1}L^2T^{-2}]$ d. $[M^{-1}L^2T^{-1}]$
- Which of the following combinations of three dimensionally different physical quantities P, Q and R can never be a meaningful quantity?
a. PQ/R b. PQ/R c. R/(PQ) d. (PR-Q)/QR
- Which of the following statements is false?
a. Having magnitude and direction is sufficient to be a vector quantity
b. Distance is a scalar quantity but displacement is a vector quantity
c. Momentum, force and velocity are vector quantities
d. Mass, speed and energy are scalar quantities.
- At what temperature do the Fahrenheit and Celsius thermometer shows the same reading?
a. -40°F b. 212°F c. 32°F d. -81°F
- Which of the following value of charge is not possible?
a. $1.6 \times 10^{-18}\text{C}$ b. $3.2 \times 10^{-18}\text{C}$
c. $3.6 \times 10^{-20}\text{C}$ d. 0.5C
- If a body is charged by rubbing, its mass
a. increases slightly b. decreases slightly
c. remains same d. may increase or decrease slightly
- When a glass rod is rubbed with woollen cloth
a. Negative charges produced on wool, but glass rod remains neutral
b. equal and opposite charges are produced on the both

- equal and similar charges are produced on the both
- positive charge is produced on the glass rod but no charge on wool.

GROUP 'B' [4x5=20]

Short answer questions

- Name any two physical quantities which have same dimensions.
- A body moving through air at high-speed V experiences a retarding force F given by $F = KAv^p$ where A is the surface area of body, p is the density of the air and K is numerical constant. Deduce the value of p.
- Taking force, length and time to be fundamental quantities. Find the dimensional formula of velocity.
- State parallelogram law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors.
- The magnitude of two vectors is equal. Show that their resultant divides the angle between them.
- Should a quantity having magnitude and direction be necessarily a vector? Explain with an example.
- Can two vectors of same magnitude have resultant equal to either of them? Explain.
- If $y = axbt^2$, Where y is the distance and t is the time. What is the dimension and unit of c.
- How can you charge a body positively by induction? Explain.
- Explain the modern theory of electrification.

GROUP 'C' [1x8=8]

Long answer questions.

- Define absolute zero temperature.
- Among two thermometers, one with spherical bulb and another with cylindrical bulb, which will respond quickly to temperature change? Why?
- A thermometer is not calibrated correctly. So it reads melting point of ice as -5°C . What is the temperature of boiling of water on the same scale if this thermometer reads 70°C instead of 60°C ?
- Discuss thermal equilibrium as a working principle of mercury thermometer.

Best of Luck

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Attempts all the questions.

GROUP 'A' MCQ [7x1=7]

- The dimensions of universal gravitational constant are
a. $[M^{-1}L^2T^{-2}]$ b. $[M^{-1}L^2T^{-1}]$
c. $[M^{-1}L^2T^{-3}]$ d. $[M^{-1}L^2T^{-2}]$
- Which of the following combinations of three dimensionally different physical quantities P, Q and R can never be a meaningful quantity?
a. PQ/R b. PQ/R c. P-Q/R d. (PR-Q)/QR
- Which of the following statements is false?
a. A vector has only magnitude, but scalar has both magnitude and direction
b. Distance is a scalar quantity, but displacement is a vector quantity
c. Momentum, force and velocity are vector quantities
d. Mass, speed and energy are scalar quantities.
- If change in temperature in Celsius scale is 45°C , the temperature change in Fahrenheit scale will be
a. 45°F b. 212°F c. 32°F d. 81°F
- Which of the following values is not possible?
a. $1.6 \times 10^{-18}\text{C}$ b. $3.2 \times 10^{-18}\text{C}$
c. $3.6 \times 10^{-20}\text{C}$ d. $1.6 \times 10^{-17}\text{C}$
- On charging by conduction mass of the body may
a. increase b. decrease
c. remains same d. increase or decrease
- When a glass rod is rubbed with silk
a. Negative charges produced on silk but glass rod remains neutral
b. equal and opposite charges are produced on both
c. equal and similar charges are produced on both
d. positive charge is produced on the glass rod but no charge on silk.

GROUP 'B' [4x5=20]

Short answer questions

- Name a physical quantity which have unit but not dimension.
- A body moving through air at high-speed V experiences a retarding force F given by $F = KAv^p$ where A is the surface area of body, p is the density of the air and K is numerical constant. Deduce the value of p.

dimensional formula of Power.

- State triangle law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors.
- What is the angle between a 2N force and 3N force so that their sum is 4N?
- Should a quantity having magnitude and direction be necessarily a vector? Explain with an example.
- Can two vectors of same magnitude have resultant equal to either of them? Explain.
- Show that work done, and energy have same dimensions.
- How can you charge a body positively by induction? Explain.
- Explain the meaning of conservation of charge.
- Iron chains touching to the ground are suspended from a vehicle carrying inflammable materials, why?

GROUP 'C' [1x8=8]

Long answer question.

- What is the temperature of a vacuum?
- What do you mean by thermal equilibrium? Why are object in contact come in thermal equilibrium?
- If a Fahrenheit thermometer reads 95° when the standard centigrade scale thermometer reads 37° , what is the error in the Fahrenheit thermometer?
- Explain the molecular concept of thermal energy.

Best of luck

उत्तीर्णादिक : १४

पूर्व वैमानिक परीक्षा

Source: U.S. Census Bureau, *Marriage, Divorce, Remarriage in the 1990s*, p. 10.

- शोरबा शीशंग बाहु चौबी लगायत इजनीं पुरकार एगुम सम्मान पनि पाएत गरिसकेकी हुन् । सम्पन्न पनि विश्वविद्यालयले उनलाई मायापत्र विद्याभितर उठाइबाट समेत सम्मान गरिसकेको छ । नेपाल सरकारले र एक हजारौं यी लाखको लागतमा कडाभाहीमा आवास निर्माण लागेको उनलाई सम्मान गरेको छ । उनी आँखि पनि साहित्य विज्ञानमा पूर्णतः समर्पित हुन् । यसो भए, अलक थर पनि मझुप्रे सबै साहित्यिको विज्ञानमा विरलतर समर्पित भइरहने भयम् । नेपाली साहित्यकी विरोध हुन हन् ।

क) भयमककमारी छे छे गर्न सञ्चिदनन् ?

६. विश्वको कतितांश पट्टी सोधिएका प्रश्नको उत्तर दिनुहोस् :

कून पहाइले कून खोलाले तिन्ही गांठ छेक्यो र
वीर पुढो। कून भोटीले तिन्ही यात्रा रोकेल्यो र
मरुवाको भी वेश तिन्ही कून आकाशले बाँध सको
पौरखले रानी नेपास। पहाड तराई जुटन सको।

पानाङ्क

- अ: नेपालीहरूको कस्ता वीरहरूले नेपालको रचना भएको हो ?
 ज: कर्णनाथको बाइलामा हाहा वीर पुर्खाको वीर्य गाथाको वर्णन गर्नुहोस् ।
 ७. वीर पुर्खाहरूले नेपालको इतिहास निर्माण दिएको योगदानको बारेमा वीर पुर्खा कतिवटा बाइलामा समीक्षा गर्नुहोस् ।
 ८. विद्वज्ज मध्ये कुनै एक शीर्षकमा २५० शब्दसम्म नपढाई निम्नमा लेख्नुहोस् ।
 क: नेपाल न पढेन
 ख: यम न म्हेरोज्ज्वर
 ग: मेरो देश : मेरो वीर

सकलताकी शुभकामना ॥

१. पिष्टकः बर्गैरुक्तो उष्णारस इत्यलं प्रोक्षत्वा आधृतः
 २. राशे ग्रुपुलैश्च वनस्पिण्डैः
 ३. नीचान् मनुष्यान् कृत्वा नीचानि बाह्यं गच्छेत्, अजगदं
 ४. गच्छेत्तु यदा यावच्छीमन्मिमांसामासां मृगं मयागच्छेत्
 ५. वनस्पिण्डैः रेखाङ्कनं ग्रीष्मकः शब्दकोऽपराधं विधानतः
 ६. समकः भागो मत्तांशं धेनुकृत्वा मोक्षेत्, उपले सोमेकः
 ७. विष्टकः शब्दकृत्वा बह्वक्षरमा रूपान्तरं वनस्पिण्डैः
 ८. तर्जयिष्ये मित्रिणीं हनुवन्तः, उपले हनुवतो तारिणः
 ९. उष्मांति भागो जुः।
 १०. पिष्टको वनस्पिण्डे पक्षी सोमेकः वस्तुगतं व्रतको मित्रो
 ११. औमत मास्ते धमत्तं हनुं वदेत्, उपको भुजः वचनं च क
 १२. नृत्तुमीर्षीं वराहं नै दुर्द्विः, अमत्तं च वराहका विष्ट मि
 १३. रश्च रं तिनकः सोमीर्षिण्यमत्तः वचनं च कर्मा जतिः ल
 १४. त्वमका लोति ऊ मष्ट चिनकः मष्ट वक्ता च मष्ट ल
 १५. मष्टः च मास्तिवकारका माका रति सोर्षी माषी, मास्ति
 १६. मष्टा मुदु हनुहः त्वयी मुदु पीठा मया तर आश्रु उन्मय
 १७. मयमका लः ऊ केनैत मुदु चिन्तः, केनैत मुदु भाव
 १८. रजसो जति जति मुदु दिपो जति जति हनुहः औत मष्ट
 १९. मया चिन्तः त्वत्पनमाह आको त्वयी एउठा मुदु विष्ट
 २०. मुदो इत्युपयोगः सः, अकः केनो इत्युपयोगः सः, उ
 २१. मुदु पीठे चिन्तः, त्वयी मुदु पीठे मया चरः, त्वयी मु
 २२. जति पीठे मया चरः, मैते वक्ता दुःखमका
 २३. सतिवमा विष्ट, मैते मास्तिव नै माम कर्ते, मैते मास्ति
 २४. वराहकृतापो जतिवमा त्वयी विष्टकः कर्माहः त्वरे, मैते मा
 २५. सतिवकर्मिनीं गति तिनः सोर्षीं तारिणः मास्तिवकर्म
 २६. नीचकः मयभन्ना अमत्तं च मयभन्ना आत्मीयः मिते हनु
 २७. त्वयि वचनं वरिष्ठाज्ज उले आत्मीयः, मयमे माह
 २८. त्वय वचनं सोर्षी आत्मीयः मुद्वन्तः, त्वी वक्ति जति

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' MORNING (SET-B) F.M: 35
Time: 1.5 hrs. Subject: Physics [1011] P.M: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP 'A' MCQ [7x1=7]

Multiple choice questions

1. The dimensions of Power are

- a. $[ML^2T^{-2}]$ b. $[M^{-1}L^2T^{-2}]$
c. $[M^{-1}L^2T^{-1}]$ d. $[M^{-1}L^2T^{-3}]$

2. Which of the following combinations of three dimensionally different physical quantities P, Q and R can never be a meaningful quantity?

- a. PQ-R b. PQ/R c. R/(P-Q) d. (PR-Q)/QR

3. Which of the following statements is false?

- a. Having magnitude and direction is sufficient to be a vector quantity
b. Distance is a scalar quantity but displacement is a vector quantity
c. Momentum, force and velocity are vector quantities
d. Mass, speed and energy are scalar quantities.

4. At what temperature do the Fahrenheit and Celsius thermometer shows the same reading?

- a. -40°F b. 212°F c. 32°F d. -81°F

5. Which of the following value of charge is not possible?

- a. $1.6 \times 10^{-18}\text{C}$ b. $3.2 \times 10^{-18}\text{C}$
c. $3.6 \times 10^{-20}\text{C}$ d. 0.5C

6. If a body is charged by rubbing, its mass

- a. Increases slightly b. Decreases slightly
c. remains same d. May increase or decrease slightly

7. When a glass rod is rubbed with woollen cloth

- a. Negative charges produced on wool, but glass rod remains neutral
b. equal and opposite charges are produced on the both

- c. equal and similar charges are produced on the both
d. positive charge is produced on the glass rod but no charge on wool

GROUP 'B' [4x5=20]

Short answer questions

8. a. Name any two physical quantities which have same dimensions.
b. A body moving through air at high-speed V experiences a retarding force F given by $F = K\rho V^2$ where A is the surface area of body, ρ is the density of the air and K is numerical constant. Deduce the value of x.
c. Taking force, length and time to be fundamental quantities. Find the dimensional formula of velocity.
9. a. State parallelogram law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors.
b. The magnitude of two vectors is equal. Show that their resultant divides the angle between them.
10. a. Should a quantity having magnitude and direction be necessarily a vector? Explain with an example.
b. Can two vectors of same magnitude have resultant equal to either of them? Explain.
c. If $y = axbt^2$, Where y is the distance and t is the time. What is the dimension and unit of c.
11. a. How can you charge a body positively by induction? Explain.
b. Explain the modern theory of electrification.

GROUP 'C' [1x8=8]

Long answer questions.

12. a. Define absolute zero temperature.
b. Among two thermometers, one with spherical bulb and another with cylindrical bulb, which will respond quickly to temperature change? Why?
c. A thermometer is not calibrated correctly. So it reads melting point of ice as -5°C . What is the temperature of boiling of water on the same scale if this thermometer reads 70°C instead of 60°C ?
d. Discuss thermal equilibrium as a working principle of mercury thermometer.

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL
PRE - TERM EXAMINATION 2081

Class: XI 'Science' MORNING (SET-A) F.M: 35
Time: 1.5 hrs. Subject: Physics [1011] P.M: 14
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP 'A' MCQ [7x1=7]

1. The dimensions of universal gravitational constant are
a. $[M^{-1}L^2T^{-2}]$ b. $[M^{-1}L^2T^{-1}]$
c. $[M^{-1}L^2T^{-3}]$ d. $[M^{-1}L^2T^{-2}]$
2. Which of the following combinations of three dimensionally different physical quantities P, Q and R can never be a meaningful quantity?
a. PQ-R b. PQ/R c. P-Q/R d. (PR-Q)/QR
3. Which of the following statements is false?
a. A vector has only magnitude, but scalar has both magnitude and direction
b. Distance is a scalar quantity, but displacement is a vector quantity
c. Momentum, force and velocity are vector quantities
d. Mass, speed and energy are scalar quantities.
4. If change in temperature in Celsius scale is 45°C , the temperature change in Fahrenheit scale will be
a. 45°F b. 212°F c. 32°F d. 81°F
5. Which of the following values is not possible?
a. $1.6 \times 10^{-18}\text{C}$ b. $3.2 \times 10^{-18}\text{C}$
c. $3.6 \times 10^{-20}\text{C}$ d. $1.6 \times 10^{-17}\text{C}$
6. On charging by conduction mass of the body may
a. increase b. Decrease
c. remains same d. increase or decrease
7. When a glass rod is rubbed with silk
a. Negative charges produced on silk but glass rod remains neutral
b. equal and opposite charges are produced on both
c. equal and similar charges are produced on both
d. positive charge is produced on the glass rod but no charge on silk.

GROUP 'B' [4x5=20]

Short answer questions

- a. Name a physical quantity which have unit but not dimension. [1]
b. A body moving through air at high-speed V experiences a retarding force F given by $F = K\rho V^2$ where A is the surface area of body, ρ is the density of the air and K is numerical constant. Deduce the value of x. [7]

dimensional formula of Power.

9. a) State triangle law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors. [3]
b) What is the angle between a 2N force and 3N force so that their sum is 4N? [2]
10. a) Should a quantity having magnitude and direction be necessarily a vector? Explain with an example. [1]
b) Can two vectors of same magnitude have resultant equal to either of them? Explain. [2]
c) Show that work done, and energy have same dimensions. [2]
11. a) How can you charge a body positively by induction? Explain. [3]
b) Explain the meaning of conservation of charge. [3]
c) Iron chains touching to the ground are suspended from a vehicle carrying inflammable materials, why? [1]

GROUP 'C' [1x8=8]

Long answer question.

12. a) What is the temperature of a vacuum? [1]
b) What do you mean by thermal equilibrium? Why are object in contact come in thermal equilibrium? [2]
c) If a Fahrenheit thermometer reads 95° when the standard centigrade scale thermometer reads 37° , what is the error in the Fahrenheit thermometer? [3]
d) Explain the molecular concept of thermal energy. [2]

Best of luck

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' Day Shift SET-A F.M.: 35
Time: 1:30 hrs. Subject: Mathematics (0071) P.M.: 14

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [5x1=5]

Rewrite the correct options of each question in your answer sheet.

- The statement $\sim (p \wedge q)$ is logically equivalent to
a) $\sim p \wedge q$ b) $\sim p \vee q$
c) $\sim p \vee \sim q$ d) $p \vee \sim q$
- The sentence $x-3=5$ is
a) True statement b) False statement
c) Compound statement d) Not a statement
- If the product of roots of the equation $mx^2 + (2m-1)x + 6x = 0$ is -2 then what is the value of m ?
a) 4 b) -3 c) $\frac{1}{4}$ d) $-\frac{1}{4}$
- Both roots of the equations $ax^2 + bx + c = 0$ are zero if
a) $b=0$ b) $c=0$
c) $b=0$ and $c=0$ d) $a=b$
- What is the value of $\lim_{x \rightarrow 0} \frac{x^2}{x^2}$?
a) 0 b) $\frac{1}{2}$ c) ∞ d) $-\infty$
- What is the value of $\lim_{x \rightarrow 0} \sqrt{x} = ?$
a) $\sqrt{5}$ b) 5 c) undefined d) $1/\sqrt{5}$
- What is the value of $\lim_{x \rightarrow 0} \frac{x^2 + 4x + 7}{x + 4}$?
a) $\frac{1}{4}$ b) 4 c) 2 d) $\frac{1}{2}$

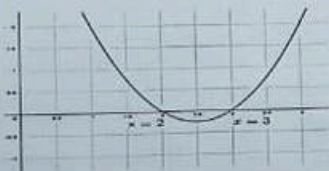
GROUP-B [5x5=25]

- Define Contradiction with example. [1]
- Write the negation of "If $5 < 6$ implies that $7 > 11$ then 8 is odd". [2]
- Construct truth table for following statement and comment the result
 $(p \vee q) \vee (\sim p)$. [2]
- Under what condition roots of quadratic equation are real and distinct? Also, explain it geometrically with example. [3]

- Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have negative roots. [1]
- If α and β are roots of $ax^2 + bx + c = 0$. Find the quadratic equation whose roots are $\alpha^2 + \beta^2$, $\alpha^{-2} + \beta^{-2}$. [2]
- a. Define limit of a function at a point. [1]
b. Evaluate $\lim_{x \rightarrow 0} \frac{x^2 - 2x}{x - 5}$
(i) Using Table
(ii) Using Formula
(iii) Is it possible to find $\lim_{x \rightarrow 0} (5)^x$ explain. [3]

GROUP-C [8x1=8]

- a. Identify the hypothesis and conclusion for "Given that 2 is an even number then 5 is a prime number".
b. From following graph.



- find the quadratic equation. [1]
- If one root of a quadratic equation is $5 + \sqrt{3}$ then find the equation. [1]
- Evaluate:
i) $\lim_{x \rightarrow -\infty} \frac{x^2 - 3}{x^2 + 2x - 3}$
ii) $\lim_{x \rightarrow 0} \frac{x^2 + 3x - 2}{x^2 - 4x + 7}$
iii) If $\lim_{x \rightarrow 0} \frac{ax^2 - 8x + 2}{3x^2 + 4x + 7} = \frac{13}{1}$ then find the value of a . [3]

The End

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' Day Shift SET-A F.M.: 35
Time: 1.5 hrs. Subject: Physics (1011) P.M.: 14
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [7x1=7]

Multiple choice questions

Assuming that the mass m of the largest stone that can be moved by a flowing river depends upon the velocity v of water, its density ρ and the acceleration due to gravity g . Then m is directly proportional to
a. v^1 b. v^2 c. v^3 d. v^4

Which of the following combinations of three dimensionally different physical quantities P , Q and R can never be a meaningful quantity?

- a. $PQ + R$ b. PQ/R c. $R/(P+Q)$ d. $(PR \cdot Q^2)/QR$

Which of the following is the scalar quantity?

- a. Displacement b. Electrostatic force
c. Acceleration d. Work

At what temperature Celsius scales reading double the Fahrenheit reading?

- a. 12.30° b. -24.60° c. 24.60° d. -12.30°

How many electrons make up a charge of 4×10^{-8} C?

- a. 2.50×10^{16} b. 4.46×10^{14}
c. 2.50×10^{18} d. 4.46×10^8

The soap bubble is given a positive charge, then its radius

- a. decreases
b. increases
c. remains unchanged
d. decreases or increases on the basis of magnitude of charge.

A conducting wire is connected between two identical conducting spheres having a charge of $-5C$ and $+1C$ respectively. Find out the new charge on each sphere.

- a. $-4C$ b. $+4C$ c. $-2C$ d. $+2C$

GROUP-B [4x5=20]

Answer question

- State principle of homogeneity. [1]
- Test dimensionally if the formula $t = 2\pi \sqrt{\frac{m}{F}}$ may be correct, where t is time period, m is mass, F is force and x is distance. [2]

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

- State triangle law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors. [3]
- Two vectors have equal magnitudes, and their resultant also has the same magnitude. What is the angle between the two vectors? [2]
- Should a quantity having magnitude and direction be necessarily a vector? Explain with an example. [1]
- Can two vectors of same magnitude have resultant equal to either of them? Explain. [2]
- If $y = a + bt + ct^2$, Where y is the distance and t is the time. What is the dimension and unit of c . [2]
- How can you charge a body negatively by induction? Explain. [3]
- Explain the meaning of the quantization of charge. [3]
- Does charge by friction follow conservation of charge? [3]

GROUP-C [1x8=8]

Long answer questions.

- Define temperature on the basis of zeroth law of thermodynamics. [1]
a) If two bodies are in thermal equilibrium in one frame, will they be in thermal equilibrium in all frames? Explain. [2]
- The steam point and ice point of a mercury thermometer are marked as 80° and 20° respectively. What will be the temperature in centigrade scale when this thermometer reads 52° ? [3]
- Write a short note about the absolute scale of temperature. [2]

Best of Luck



KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION 2082

Date: XI 'Science'

Time: 1:30 hrs.

MORNING SHIFT 'SET-A'

Subject: Chemistry [3011]

F.M: 37

P.M: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS

- $B_2S + H_2O_2 \rightarrow S + H_2O$ shows
 - Reducing action of H_2O_2
 - Oxidizing action of H_2O_2
 - Catalytic action of H_2O_2
 - Oxidizing action of H_2S
- Strongest oxidizing agent among the following is
 - $HClO$
 - $HClO_2$
 - $HClO_3$
 - $HClO_4$
- Oxidation state and valency of carbon in CH_2Cl_2 is
 - 0, 4
 - 0, 2
 - +4, 0
 - 4, 0
- Absolute zero is the temperature equal to
 - Room temperature
 - $273^\circ C$
 - $-273.16^\circ C$
 - $273.16^\circ C$
- The curve plotted between P and V at constant temperature is called
 - Isotherm
 - Isobar
 - Isochrome
 - Isotope
- Which non-metal is most active?
 - F_2
 - N_2
 - O_2
 - S
- Which of the following species has the highest electron affinity?
 - F^-
 - O
 - O^-
 - Na^+
- The size of following species increase in the order
 - $Mg^{+2} < Na^+ < F^- < Al^{+3}$
 - $F^- < Al < Na^+ < Mg^{+2}$
 - $Al < Mg^{+2} < F^- < Na^+$
 - $Na^+ < Al < F^- < Mg^{+2}$
- The density of NH_3 at $30^\circ C$ and 5 atm pressure is
 - 2.30 gm/lit
 - 3.42 gm/lit
 - 23 gm/lit
 - 30 gm/lit

GROUP - 'B' [4×3=12]

SHORT ANSWER QUESTIONS

- State Modern Periodic Law. [1]
 - Write the main features of Modern Periodic Table. [4]
- Define the terms:
 - Screening Effect
 - Effective Nuclear Charge
 [1+1]
 - Define ionisation energy. Why is the ionisation energy of nitrogen higher than that of Oxygen? [1+2]
- Derive combined gas equation. [2]
 - Find the value of R in C.G.S. unit. [2]
 - What is significance of R? [1]
- Volume of moist gas at $23^\circ C$ and 800 mm Hg pressure is 100 cc. Calculate the aqueous tension at $23^\circ C$ if the volume occupied by gas at NTP is 94.53 cc. [3]
 - Write any two applications of Boyle's Law. [2]

GROUP - 'C' [1×8=8]

LONG ANSWER QUESTION

- Differentiate between oxidation number and valency. [2]
 - Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent in the reaction,

$$KMnO_4 + KCl + H_2SO_4 \rightarrow MnSO_4 + K_2SO_4 + H_2O + Cl_2$$
 [2]
 - Justify that, " SO_2 acts as both oxidizing agent as well as reducing agent". [2]
 - Calculate the oxidation number of underlined species.
 - $Ni(CO)_4$
 - $Cr(NH_4)_2(SO_4)_3$
 - NH_4NO_3
 - SO_4^{2-}
 [2]

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' (MORNING) SET-A F.M.: 35
Time: 1:30 hrs. Subject: Chemistry [3011] P.M.: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [5x1=5]

- Which of the following species has the highest electron affinity?
(a) F^- (b) O (c) O^{2-} (d) Na^+
- Pick out the isoelectronic structures from the following:
(i) CH_4 (ii) H_2O^+ (iii) NH_3 (iv) CN_2
(a) (i)&(ii) (b) (i)&(iv) (c) (ii)&(iii) (d) (iii), (iii) & (iv)
- The gas law giving relationship between volume and temperature of a gas is
(a) Boyle's law (b) Charles's law (c) Graham's law (d) Dalton's law
- Oxidation number of phosphorus ranges from
(a) -3 to +5 (b) +5 to -5 (c) -2 to +6 (d) +2 to -6
- The reaction, $H_2S + H_2O_2 \rightarrow S + H_2O$ shows,
(a) Reducing action of H_2O_2 (b) Oxidizing action of H_2O_2
(c) Catalytic action of H_2O_2 (d) Oxidizing action of H_2S

GROUP 'B' [5x2=10]

- Short Questions.
- Show with examples that all reactions are not redox reactions. [2]
 - Distinguish between valency and oxidation number. [2]
 - Identify the oxidation number of S in H_2SO_5 . [1]
 - Balance the following redox reaction by oxidation number method.
 $K_2Cr_2O_7 + FeSO_4 + H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + Fe_2(SO_4)_3 + H_2O$ [1]
 - Identify oxidizing and reducing agent in above reaction. [1]
 - Define absolute zero temperature according to Charles's law with the help of suitable graph. [2]
 - Derive the ideal gas equation. [2]
 - What is the application of Boyle's law? [1]
 - An open vessel at $27^\circ C$ is heated until three-fifths of the air in it has been expelled. Assuming that the volume of vessel remains constant, find the temperature to which the vessel has been heated? [3]
 - Plot the graph for
(i) Volume Vs 1/pressure (ii) Pressure X volume Vs Pressure [1+1]
 - Define the terms (i) Screening Effect (ii) Effective Nuclear Charge. [1+1]
 - Define electron affinity. Why is the electron affinity of chlorine greater than that of Fluorine? [1+2]
 - State modern periodic law. Write the advantages of modern periodic table. [1+4]

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' (MORNING) SET-B F.M.: 35
Time: 1:30 hrs. Subject: Chemistry [3011] P.M.: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [5x1=5]

- The process requiring the absorption of energy is
(a) $F \rightarrow F^-$ (b) $Cl \rightarrow Cl^-$ (c) $O \rightarrow O^{2-}$ (d) $H \rightarrow H^-$
- Which has the highest electronegativity?
(a) C (b) Mg (c) O (d) S
- Oxidation state of oxygen atom in potassium superoxide is
(a) $-\frac{1}{2}$ (b) -1 (c) -2 (d) 0
- A solution of SO_2 in water reacts with H_2S precipitating sulphur. Here, SO_2 acts as
(a) an oxidizing agent (b) A reducing agent
(c) an acid (d) a catalyst
- A gas is initially at 1 atm pressure. To compress it to $\frac{1}{8}$ th of its initial volume, pressure to be applied is
(a) 1 atm (b) 2 atm (c) 4 atm (d) $\frac{1}{4}$ atm

GROUP 'B' [5x2=10]

- Calculate the oxidation number of Cr in CrO_5 with structure. [2]
- HNO_3 acts as oxidizing and agent, why? [2]
- Identify oxidation and reduction reaction as well as oxidant and reductant in the given reaction.
 $K_2Cr_2O_7 + FeSO_4 + H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + Fe_2(SO_4)_3 + H_2O$
- Balance the following redox reaction by oxidation number method.
 $KBr + MnO_2 + H_2SO_4 \rightarrow KHSO_4 + MnSO_4 + Br_2 + H_2O$
- Give example for maximum and minimum oxidation number of Cl. [2]
- Derive combined gas equation. [2]
- Find the value of 'n' in SI unit. [1]
- Give the application of Charles's law. [1]
- A balloon can hold 1000 cc of air before bursting. The balloon can hold 975 cc of air at $5^\circ C$. Will it burst when it is taken into a house at $25^\circ C$? Assume that the pressure of the gas in the balloon remains constant. [1]
- Explain the nature of curve obtained during isothermal conditions.
(a) Compare the ionic size of (i) Na and Na^+ (b) Cl and Cl^- [1+1]
- Explain the difference between electron affinity & electronegativity. [2]
- Define ionization potential. What are the various factors that affect ionization potential? Explain. [1+2]
- Why is the ionization potential of Be higher than that of B? [2]

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' Day Shift, Set-B F.M.: 35
Time: 1:30 hrs. Subject: Mathematics [0671] P.M.: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [5x1=5]

- Write the correct options of each question in your answer sheet.
- The statement $\sim (p \vee q)$ is logically equivalent to
(a) $\sim p \wedge \sim q$ (b) $\sim p \wedge q$
(c) $\sim p \vee \sim q$ (d) $\sim p \vee q$
- The value of $\lim_{x \rightarrow 0} \frac{x^2 + 12x^3}{x + 6}$ is
(a) $\frac{1}{6}$ (b) 2 (c) 4 (d) 0
- The sentence $x+2=5$ is
(a) True statement (b) False statement
(c) Compound statement (d) Not a statement
- Both roots of the equation $ax^2 + bx + c = 0$ are zero if
(a) $b = 0$ (b) $c = 0$
(c) $b = 0$ and $c = 0$ (d) $a \neq 0$
- The value of $\lim_{x \rightarrow 0} \frac{x^2 - 4}{x^2}$ is
(a) 0 (b) $\frac{1}{4}$ (c) ∞ (d) $-\infty$
- The value of $\lim_{x \rightarrow 0} 0$ is
(a) 12 (b) 0 (c) undefined (d) 5

- If the product of roots of the equation $mx^2 + (2m+1)x - 6 = 0$ is -2 then what is the value of m?
(a) 3 (b) -3 (c) $\frac{1}{3}$ (d) $-\frac{1}{3}$

GROUP 'B' [5x2=10]

- Define contradiction with example. [1]
- Write the negation of "If $5 < 6$ implies that $7 > 11$ then 8 is odd". [2]
- Construct truth table for following statement and comment on the result $(p \vee q) \vee (\sim p)$. [2]
- Under what condition roots of the quadratic equation are real and distinct? Also, explain it geometrically with an example. [2]

b. Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have negative roots. [2]

10. If α and β are roots of $ax^2 + bx + c = 0$, find the quadratic equation whose roots are $\alpha^2 + \beta^2$, $\alpha^{-1} + \beta^{-1}$. [2]

11. a. Define limit of a function $f(x)$ at a point a . [1]

b. Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$. [2]

(i) Using Table [1]

(ii) Using Formula [1]

(iii) Is it possible to find $f(1)$? Explain. [1]

GROUP 'C' [3x1=3]

12. a) Evaluate $\lim_{x \rightarrow 0} \frac{x^2 - 3}{x^2 + 3x - 36}$. [2]

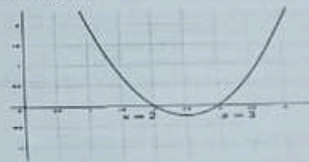
(i) $\lim_{x \rightarrow 0} \frac{x^2 - 3}{x^2 + 3x - 36}$ [1]

(ii) $\lim_{x \rightarrow 0} \frac{x^2 - 3}{x^2 + 3x - 36}$ [1]

(iii) If $\lim_{x \rightarrow 0} \frac{x^2 - 3}{x^2 + 3x - 36} = \frac{1}{3}$ then find the value of a . [2]

b) Identify the hypothesis and conclusion for "Given that 2 is an even number then 5 is a prime number". [1]

c) From following graph.



find the quadratic equation. [2]

d) If one root of a quadratic equation is $5 - 4\sqrt{3}$, find the equation. [1]

"The End"



KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2082

Class: XI 'Science'

Time: 1:30 hrs.

MORNING SHIFT 'SET-B'

Subject: Chemistry [3011]

F.M: 37

P.M: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS

- The oxide which cannot act as reducing agent is
a) NO_2 b) CO_2 c) SO_2 d) ClO_2
- Oxygen has positive oxidation state in
a) H_2O_2 b) Cl_2O c) F_2O d) NaOCl
- Oxidation number of two chlorine atoms in bleaching powder (CaOCl_2) are
a) -1, -1 b) +1, +1 c) +1, -1 d) 0, -1
- If both temperature and volume of a gas are doubled, the pressure
a) is doubled b) remains unchanged
c) is reduced to half d) is tripled
- One litre of an unknown gas weighs 1.25 gm at NTP. Which of the following gas corresponds above data?
a) NO_2 b) N_2 c) CO_2 d) O_2
- The outermost electronic configuration of the most electronegative element is
a) ns^2np^3 b) ns^2np^4 c) ns^2np^5 d) ns^2np^6
- Correct order of electron affinities is
a) $\text{Cl} > \text{Br} > \text{I} > \text{F}$ b) $\text{Cl} > \text{F} > \text{Br} < \text{I}$
c) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ d) $\text{Cl} > \text{F} > \text{Br} > \text{I}$
- Pick the isoelectronic structures from the following
i) CH_3^+ ii) H_2O^+ iii) NH_3 iv) CH_3^-
a) i) & ii) b) iii) & iv) c) i) & iii) d) ii), iii) & iv)
- A gas which perfectly obeys gas laws at all temperature is called
a) Noble gas b) permanent gas c) Inert gas d) Ideal gas

GROUP - 'B' [4×5=20]

SHORT ANSWER QUESTIONS

- a) Define periodicity. [1]
b) Describe Bohr's classification of elements. [4]
- a) Define electron affinity. [1]
b) Compare the size of Na^+ , Mg^{2+} & Al^{3+} with reasons. [2]
c) Why is electron affinity of chlorine higher than fluorine? [2]
- a) State Boyle's law and hence derive its expression. [1+1]
b) Graphically represent all the isotherms. [3]
- a) Define absolute zero with graph. [1+1]
b) A flask of 2.5 litres capacity can withstand a pressure of 14 atm and is filled with CO_2 gas at 27°C and 760 mm Hg pressure. At what temperature will it explode on heating? [3]

GROUP - 'C' [1×8=8]

LONG ANSWER QUESTION

- a) Justify the statement, "Oxidation and reduction occurs simultaneously". [2]
b) Define autoredox reaction with example. [2]
c) Calculate the oxidation number of underlined species.
i) $\text{K}_4[\underline{\text{Fe}}(\text{CN})_6]$ ii) $\underline{\text{C}}_6\text{H}_{12}\text{O}_6$
iii) $\underline{\text{N}}\text{H}_4\text{NO}_3$ iv) $\underline{\text{Fe}}_2\text{O}_4$ [2]
d) Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent. [2]
 $\text{K}_2\text{Cr}_2\text{O}_7 + \text{FeSO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$

Best of Luck

Class: XI 'Science'
Time: 1:30 hrs.
Subject: Mathematics (0071)
Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [5x1=5]

1. What is the contrapositive of $p \Rightarrow q$?
a) $p \Rightarrow q$ b) $q \Rightarrow p$ c) $\neg p \Rightarrow \neg q$ d) $p \Rightarrow \neg q$
2. What is the negation of $p \Rightarrow q$?
a) $p \vee q$ b) $p \wedge \neg q$ c) $\neg p \wedge q$ d) $\neg p \vee q$
3. If the roots of the equation $ax^2 + bx + c = 0$ are equal in magnitude but opposite in sign then
a) $a = 0$ b) $b = 0$ c) $c = 0$ d) $a \neq b$
4. If α and β are roots of $ax^2 + bx + c = 0$, what is the value of $\alpha\beta^2 + \alpha^2\beta + \alpha\beta$?
a) $\frac{c(b+c)}{a^2}$ b) $\frac{(a+b)}{a^2}$ c) $\frac{bc}{a^2}$ d) $\frac{(a+b)}{a^2}$
5. What is the value of $\lim_{x \rightarrow 0} \frac{1}{x^2}$?
a) 0 b) $\frac{1}{0}$ c) ∞ d) $-\infty$
6. What is the value of $\lim_{x \rightarrow 0} \frac{1}{x}$?
a) 5 b) 8 c) undefined d) π
7. What is the value of $\lim_{x \rightarrow 0} \frac{x^2 + 2x}{x^2 + 1}$?
a) $\frac{1}{2}$ b) $\frac{1}{4}$ c) $\frac{1}{2}$ d) $\frac{1}{4}$

GROUP-B [5x5=25]

8. Define tautology with an example.
a) Write the negation of "3 < 4 implies that 3 > 7 then 4 is odd".
b) Construct truth table for following statement and comment on it.
c) Construct truth table for following statement and comment on it.
d) $(p \wedge q) \Rightarrow p \vee q$
9. a) Under what condition the roots of quadratic equation are real and equal? Also, interpret it geometrically.
b) Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have positive roots.
c) Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have whose roots are $\alpha^2 + \beta^2$, $\alpha - \beta$.
10. If α and β are roots of $ax^2 + bx + c = 0$, find the quadratic equation whose roots are $\alpha^2 + \beta^2$, $\alpha - \beta$.

Class: XI 'Science'
Time: 1:30 hrs.
Subject: Mathematics (0071)
Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [5x1=5]

1. What is the contrapositive of $p \Rightarrow q$?
a) $p \Rightarrow q$ b) $q \Rightarrow p$ c) $\neg p \Rightarrow \neg q$ d) $p \Rightarrow \neg q$
2. What is the negation of $p \Rightarrow q$?
a) $p \vee q$ b) $p \wedge \neg q$ c) $\neg p \wedge q$ d) $\neg p \vee q$
3. If the roots of the equation $ax^2 + bx + c = 0$ are equal in magnitude but opposite in sign then
a) $a = 0$ b) $b = 0$ c) $c = 0$ d) $a \neq b$
4. If α and β are roots of $ax^2 + bx + c = 0$, what is the value of $\alpha\beta^2 + \alpha^2\beta - \alpha\beta$?
a) $\frac{c(b+c)}{a^2}$ b) $\frac{(a+b)}{a^2}$ c) $\frac{bc}{a^2}$ d) $\frac{(a+b)}{a^2}$
5. What is the value of $\lim_{x \rightarrow 0} \frac{1}{x^2}$?
a) 0 b) $\frac{1}{0}$ c) ∞ d) $-\infty$
6. What is the value of $\lim_{x \rightarrow 0} \frac{1}{x}$?
a) 5 b) 8 c) undefined d) π
7. What is the value of $\lim_{x \rightarrow 0} \frac{x^2 + 2x}{x^2 + 1}$?
a) $\frac{1}{2}$ b) $\frac{1}{4}$ c) 2 d) 1

GROUP-B [5x5=25]

8. a) Write the truth value of "3 is odd and 4 is even". Also, write its negation.
b) Construct truth table for the following statement and comment on it.
c) Construct truth table for the following statement and comment on it.
d) $(p \wedge q) \wedge (p \Rightarrow q)$
9. a) Under what condition roots of the quadratic equation are imaginary? Also, interpret it geometrically.
b) Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have positive roots.
c) Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have whose roots are $\alpha^2 + \beta^2$, $\alpha - \beta$.
10. If α and β are roots of $ax^2 + bx + c = 0$, find the quadratic equation whose roots are $\alpha^2 + \beta^2$, $\alpha - \beta$.

a) Define limit of a function f(x) at a point x=a.

b) Evaluate $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2}$

c) Using Table

d) Using Formula

e) Is it possible to find f(2)? explain.

f) Is it possible to find f(2)? explain.

g) Identify the hypothesis and conclusion for "If B is multiple of 2 then B is even".

h) From following graph,

i) Find the quadratic equation.

j) If one root of a quadratic equation is $2 + \sqrt{3}$, then find the equation.

k) Evaluate:

l) $\lim_{x \rightarrow 0} \frac{x^2-1}{x^2+1}$

m) $\lim_{x \rightarrow 0} \frac{x^2+2x-1}{x^2+1}$

n) If $\lim_{x \rightarrow 0} \frac{x^2+2x}{x^2+1} = \frac{2}{3}$, then find the value of p.

o) The End

a) Define limit of a function f(x) at a point x=a.

b) Evaluate $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2}$

c) Using Table

d) Using Formula

e) Is it possible to find f(2)? explain.

f) Is it possible to find f(2)? explain.

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k) Evaluate:

l) $\lim_{x \rightarrow 0} \frac{x^2-1}{x^2+1}$

m) $\lim_{x \rightarrow 0} \frac{x^2+2x-1}{x^2+1}$

n) If $\lim_{x \rightarrow 0} \frac{x^2+2x}{x^2+1} = \frac{2}{3}$, then find the value of p.

o) The End

6. Choose the best options:

- a) The prefix 'dis-' goes with the word [5x1=5]
i) obey ii) possible iii) legal iv) happy
- b) The meaning of the prefix in the word 'preview' is
i) again ii) before iii) wrong iv) not
- c) The mountain is very steep. Choose the correct meaning of the word 'steep'.
i) flat ii) sloppy iii) wide iv) low
- d) If the guide words are "king – kite", which of the following will appear on that page?
i) kind ii) kitten iii) kitchen iv) keep
- e) 'Glory' comes before 'Gold' because comes before in the alphabet. (Fill in the blanks with suitable letters)
i) l/o ii) d/r iii) r/y iv) g/o

[Best Wishes]

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION- 2082

Class: XI
 Time: 1hr.

Morning Shift Set 'A'
 Subject: Computer Science [4271]

F. M.: 25
 P. M.: 10

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP – 'A' MCQ [5×1=5]

Multiple Choice Questions.

- The brain of Computer System is?
 a) ALU b) Memory c) Control Unit d) CPU
- Microprocessor was introduced in which generation of computer?
 a) Third b) Second c) Fourth d) First
- Which of the following is exclusively a sequential access storage device?
 a) Hard Disk b) Floppy Disk c) DVD d) Magnetic Tape
- Which is the speed of computer measured in?
 a) Nanoseconds b) Kilo-Seconds c) Gigahertz d) Megabytes
- Which of the following is not considered hardware?
 a) Operating System. b) CPU. c) Keyboard. d) Hard disk.

GROUP – 'B' [3×4=12]

Short Answer Questions.

- What do you mean by "Volatility"? Explain RAM and ROM with the concept and term. [1+3]
- What is a 'BUS' in computer architecture? Explain different types of 'BUS' in brief. [1+3]

'OR'

Explain cash memory and Buffer Memory in details. [2+2]

- Differentiate between hibernate and sleep mode in PC. [4]

GROUP – 'C' [1×8=8]

Long Answer Questions.

- What do you mean by computer architecture? Describe each functional components of computer architecture with typical diagram. [2+6]

"Best of Luck"

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION- 2082

Class: XI
 Time: 1hr.

Morning Shift Set 'A'
 Subject: Computer Science [4271]

F. M.: 25
 P. M.: 10

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP – 'A' MCQ [5×1=5]

Multiple Choice Questions.

- The brain of Computer System is?
 a) ALU b) Memory c) Control Unit d) CPU
- Microprocessor was introduced in which generation of computer?
 a) Third b) Second c) Fourth d) First
- Which of the following is exclusively a sequential access storage device?
 a) Hard Disk b) Floppy Disk c) DVD d) Magnetic Tape
- Which is the speed of computer measured in?
 a) Nanoseconds b) Kilo-Seconds c) Gigahertz d) Megabytes
- Which of the following is not considered hardware?
 a) Operating System. b) CPU. c) Keyboard. d) Hard disk.

GROUP – 'B' [3×4=12]

Short Answer Questions.

- What do you mean by "Volatility"? Explain RAM and ROM with the concept and term. [1+3]
- What is a 'BUS' in computer architecture? Explain different types of 'BUS' in brief. [1+3]

'OR'

Explain cash memory and Buffer Memory in details. [2+2]

- Differentiate between hibernate and sleep mode in PC. [4]

GROUP – 'C' [1×8=8]

Long Answer Questions.

- What do you mean by computer architecture? Describe each functional components of computer architecture with typical diagram. [2+6]

"Best of Luck"



1. Read the following text and do the activities that follow.

Long long ago, in the small village, people were terrified of an old abandoned mansion at the edge of the forest. Every night, strange noises echoed from inside—footsteps, whispers, and sometimes a chilling scream. Villagers first thought it was robbers or wild animals. But the weirdest thing was that the noises always happened exactly at midnight. Raj, a brave but lazy teenager, decided to investigate. Not because he wanted to explore, but because his younger sister dared him. He entered the mansion with only a flashlight and a small notebook.

At he stepped inside, the door creaked loudly behind him. Raj froze. Shadows flickered on the walls, and cold gusts of wind blew through the broken windows. Suddenly, he saw a pair of glowing eyes staring at him from the dark hallway. He quickly took a step back and realized it was a black cat—but then he heard a whisper saying, "Leave now..."

The next morning, a local historian explained that the mansion had been abandoned for decades after a family mysteriously disappeared. The noises were caused by the wind moving through broken floorboards and shutters, and the villagers' fear made the experience feel more terrifying than it really was.

From then on, people avoided the mansion at night, but some teenagers still dared each other to peek inside for a thrill.

A. Answer the following questions in a single sentence each: [5x1=5]

- What was strange about the abandoned mansion?
- Why did Raj decide to enter the mansion?
- How did the local historian explain the strange noises?
- What did Raj see first when he entered the mansion?
- What did people do after learning the truth about the mansion?

B. Fill in the blanks with suitable information from the passage. [5x1=5]

- People first thought the noises were caused by
- The noises always happened exactly at
- Raj entered the mansion with a flashlight and a small
- Raj saw a pair of glowing staring at him.
- The abandoned mansion was located at the edge of the

2. Explain with reference to the context: [5]

- Stop crying, Oh man,

Stop cursing me

And listen.

I didn't come here of my own free will.

I was invited.

Believe me, I had no choice but to visit you.

How many lives were lost

Because of me? You count.

But how you ever counted

How many have died so far

Because of you and your wars?

3. Answer the following questions in about 75 words each. [3x2=6]

- Why did the Giant knock down the high wall around his garden? (The Selfish Giant)
- What is personification? Explain the use of it in the story, The Selfish Giant
- What human activities invited Corona to the world? (Corona Soys)

4. Write a readable story using the clues given below: [8]

dog finds a piece of meat..... crosses a bridge..... sees his reflection..... thinks another dog has meat..... tries to snatch..... drops his own meat..... moral

5. Rewrite the sentences as indicated in the brackets. [3x1=3]

- He bought a car. (Put the adverb recently in the correct place.)
- She speaks English. (Put the adverb fluently in the correct place.)
- We play football. (Put the adverb hardly in the correct place.)



KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION 2082

Class XI
 Date: 1.5 hrs.
 Subjects: Mathematics Set-A
 P.M.: 37
 P.M.: 15
 Candidates are required to give their answers in their own words as far as possible. The figures in the margin indicate full marks.
 Attempt all the questions.

GROUP-A [MCQ] [4x1=4]

Write the correct options of each question in your answer sheet.

- Which of the following is the contrapositive of $p \Rightarrow \sim q$?
 A) $p \Rightarrow \sim q$ B) $q \Rightarrow p$ C) $q \Rightarrow \sim p$ D) $p \Rightarrow q$
- If $x \in (A \cup B)$, what does it mean?
 A) $x \in A$ and $x \in B$ B) $x \in A$ or $x \in B$
 C) $x \in A$ and $x \in B$ D) $x \in A$ and $x \in B$
- What is the value of $\lim_{x \rightarrow \infty} \frac{1+x^2}{4+x}$?
 A) e B) 5 C) undefined D) π
- What is the value of $\lim_{x \rightarrow \infty} \frac{1+x^2}{4+x}$?
 A) $\frac{1}{4}$ B) 4 C) 2 D) $\frac{1}{2}$

GROUP-B [5x5=25]

- Write the negation of the statement:
 "If $3 < 4$ implies that $3 > 7$ then 4 is odd".
 Also, determine its truth value. [1+1]
- Construct truth table for the statement: $(p \wedge q) \wedge (\sim p \vee \sim q)$.
 Also, what conclusion can be drawn from the truth table. [2+1]
- Define intersection of two sets. [1]
- List the various indeterminate forms that occur in limits. [1]
- Write the negation of $p \vee q$. [1]
- What is the value of $\lim_{x \rightarrow 0} \frac{x}{\sin x}$? [1]
- Define the complement of a set with an example. [1]

- State and prove De-Morgan's laws for two sets. [1+4]

- Evaluate $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$
 a) Using a table of values.
 b) Using algebraic formula.
 c) Is it possible to find $f'(3)$? Explain. [2+2+1=5]
- If $A = \{a, b, c\}$ and $B = \{a, b\}$, verify that $P(B) \subseteq P(A)$. [2]
 Evaluate $\lim_{x \rightarrow y} \frac{x^2+y^2}{x^2-y^2}$ [3]

GROUP-C [8x1=8]

- Does the order of elements in a set have any significance? Justify your answer with an example. [2]
- What is the difference between undefined and indeterminate forms in the context of limits? Justify your answer. [3]
- Give an example of a function which takes the indeterminate form $\frac{0}{0}$ but whose limit does not exist. Justify your answer. [3]

Best of Luck

Class: XI

Time: 1hr.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP - 'A' [5×1=5]

Multiple Choice Questions.

1. 4 bits equals to:

- a) 1 byte b) A nibble c) A block d) 2 nibbles

2. Which generation of computers introduced the use of integrated circuits (ICs)?

- a) Third b) Second c) Fourth d) First

3. What is the primary function of cache memory in a computer system?

- a) To run applications without needing an internet connection
b) To store the operating system permanently
c) To provide long-term, archival storage
d) To act as a temporary holding area for frequently accessed data

4. MIPS stands for?

- a) Multiple Instructions Per Second
b) Million Instructions Per Second
c) Major Instructions Per Second
d) Million Information Per Second

5. Which of the following is not considered output device?

- a) Projector b) Screen c) Speaker d) microphone

GROUP - 'B' [3×4=12]

Short Answer Questions.

1. What do you mean by Primary memory? Differentiate Between SRAM and DRAM. [1×3]

2. What is CPU? Explain different components of CPU in brief. [1×3]

'OR'

Explain Address Bus and Data Bus in details.

[2+2]

3. Imagine you are designing a brochure for your school's annual fest. Using MS word, describe the steps of creating an eye catching cover page that includes:

[4]

- ❖ A catchy title
- ❖ An image related to the theme
- ❖ A slogan of the fest
- ❖ A background color or texture that complements the theme

GROUP - 'B' [3×4=12]

Long Answer Questions.

1. What do you mean by computer memory? Describe different types of computer memory with clear diagram. [2+6]

"Best of Luck"

प्रश्नहरू :

- (क) केले नागा टुटाउँछ, केले नागा जोड्छ ?
 (अ) रिक्खले टुटाउँछ, यवालले जोड्छ । (आ) बारियाले टुटाउँछ, मायाले जोड्छ ।
 (इ) पुषाले टुटाउँछ, विमले जोड्छ । (ई) लोभले टुटाउँछ, दानले जोड्छ ।
- (ख) हजारौं वर्ष पहिले कस्तो चलन थियो ?
 (अ) खेतीपाती नगर्ने (आ) लुग्रा नलगाउने
 (इ) घर नबनाउने (ई) एकजनालाई सहयोग नगर्ने
- (ग) कौनो खानाको के भइसक्यो ?
 (अ) पोतेको खाना (आ) उमातेको खाना
 (इ) पक्काल्ल (ई) भाजी खाना
- (घ) रथका ठाउँमा के भइसक्यो ?
 (अ) रेल, मोटर, ट्याक्सी र हवाईजहाज (आ) रेल, मोटर, हवाईजहाज र रकट
 (इ) रेल, मोटर, कार र मोटरसाइकल (ई) रेल, मोटर, ट्याक्सी र रकट

५. प्रश्नको उत्तर लेख्नुहोस् ।

क. दिइएको कवितामा पढी सोधिएका प्रश्नहरूको दिनुहोस् :

(Y+Y=C)

द्विगर्धमा डोभ तिघो चिनो हिउँथिच होला
 वंशावती किनारभरि पौरखको चिनो होला
 वीर पुर्खा ! तिम्रीलाई भित्तैरिते माथै बाँध्यो
 सागर तरी समाभरि वीर गोर्खा रगत बग्यो ।

प्रश्नहरू :

- (अ) माथिको कवितामा हाप्पा पुर्खाहरूका बारेमा के भनिएको छ ?
 (आ) हामीले हाप्पा पुर्खाको गौरव कसरी जोगाउन सक्छौं ?

ख. तलको कथा पढी सोधिएका प्रश्नको उत्तर दिनुहोस् :

खोला बघेल फेर गाउँतिर साँझको रहेछ । यो खोलाते पनि गाउँलाई धुन्कै रुवायो । हुन त रुवायो माथै भन्नु खोलाप्रति अन्याय ठहर्ने । यो खोलाते गाउँलाई पार्नेको पनि छ । हाप्पो विगतको लगभग आधा उमेर यही खोलाको किन र बगरहरूमा बितेको थियो । कहिले बोरहा खाने निर्हुमा त कहिले गाईभैसी चराउने निर्हुमा । खोलाका बाढी आएको बेला हामी गाईको पुच्छर समाएर बाढी तर्ने गर्छौं जबका बस्ती बेलासा हामी खोलाको किनारमा बसेर बाढीको आनन्द लुट्छौं । तर्ने बराउने कोही मानिस आए भने हामी तीनचार जना एकजनाको हात समाउँदै बाढीमा होतन्थौं र किनारमा परिरेहको मानिसलाई तारिदिन्थौं र बस्ती कसममा खान गरी परिछनकै विशेष रसि थियो । "योतिनर कुवा छ, गए सालको वर्षामा मूल कुटेको एकदम थिनो र मिठो पानी छ ।" ऊ पानी लिएर आइसकेको थियो ।

प्रश्नहरू :

- (अ) खोलाते रुवायो भन्ने कुन यात्र गर्दा कसरी अन्याय हुन्छ ?
 (आ) "य" पात्रका खोलासँगका बालबनुपुति कस्ता छन् ?

७. वीर पुर्खाहरूले कसरी वीरता देखाए ? समीक्षात्मक उत्तर लेख्नुहोस् ।

८

सफलताको शुभकामना !!



KATHMANDU MODEL SECONDARY SCHOOL

PRE - TERM EXAMINATION 2082

Ques: XI 'Science'
MORNING SHIFT 'SET-B'
Subject: Physics [1011]

F.M: 37
P.M: 15
Time: 1.5 hrs.
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP - 'A' [6X1=6]

Multiple Choice Questions:

1. The equation of a wave is given by, $y = A \sin \left[\omega \left(\frac{x}{v} - t \right) \right]$, when ω is angular velocity x is position and v is linear velocity. The dimensional formula for k is:

- a) $[L^{-1}]$ b) $[T]$ c) $[T^{-1}]$ d) $[T^{-2}]$

2. Identify the correct statement:

- a) Velocity, displacement, acceleration, displacement are vectors
b) Weight, mass, torque and energy are scalars
c) Angular velocity, momentum and electric current are scalars
d) A vector quantity is one which has magnitude only whereas scalar quantity is one which has both magnitude and direction.

When 2×10^{13} electrons are added to a neutral metal sphere, the charge on the sphere becomes:

- a) $-16 \mu C$ b) $16 \mu C$ c) $32 \mu C$ d) $-32 \mu C$

A soap bubble is given a negative charge, then its radius:

- a) Decreases b) Increases c) Remains unchanged
d) Nothing can be predicted as information is insufficient.

The force between two charges 0.06 m apart is 5 N . If each charge is moved towards the other by 0.01 m , then the force between them will be

- a) 7.20 N b) 11.25 N c) 22.50 N d) 45.0 N

When a positively charged, conductor is connected to earth:

- a) No charge flow occurs
b) Electrons flow from conductor to earth
c) Protons flow from conductor to earth
d) Electrons flow from earth to conductor

GROUP - 'B' [5X3=15]

Short Answer Questions:

a) What are the basic rules of dimensional analysis?

[1]

b) Can we tell the unit of a physical quantity from its dimensions? What about dimensions from the unit of a physical quantity?

[1]

c) Check the correctness of the physical equations: (i) $t = 2\pi \sqrt{\frac{l}{g}}$; t is time period, l is length and g is acceleration due to gravity (ii) $T^2 = \frac{4\pi^2 a^3}{GM}$; T is time period, G is gravitational constant, M is mass and a is radius of the orbit.

[3]

d) What is meant by resultant vector.

[1]

e) Two vectors acting on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangular representation? State and explain the law involved.

[3]

f) Under what condition is the magnitude of resultant vector is zero vector?

[1]

g) a) Explain about Action of point with a demonstration? b) What do you mean by electrostatic induction? How will you give permanent positive charge to a conductor by the method of induction? Explain in detail.

[3]

GROUP - 'C' [2X8=16]

Long Answer Questions

10. a) What do you mean by dimension of a physical quantity? What is the difference between dimensions and unit of a physical quantity?

[2]

b) A body moving through air at a speed v experiences a retarding force F given by the equation $F = kAv^p$, where A is the surface area of the body, p is the density of air and k is dimensionless constant. Deduce the value of p .

[2]

c) Convert 60 J/s into a new system of unit which has 10 gm , 10 cm and 10 second as fundamental units of mass, length and time.

[2]

d) Is it possible to derive all the physical equations by the method of dimensional method? Explain.

[2]

11. a) Write the formula for Coulomb's force in vector form? What is the significance while expressing in vector form?

[2]

b) What is electrical permittivity? Write its value and unit for vacuum in SI and cgs system of unit.

[2]

c) If a body gives out 10^9 electrons per second, how much time is required to get a total charge of 1 C from it? What conclusion would you draw from the result obtained?

[2]

d) Two positive point charges are at a separation of 0.3 m repel each other with a force of 1.6 N . If the sum of two charges be 10 microcoulomb, what are their separate values?

[2]

Best of Luck

Attempts all the questions.

Multiple Choice Questions:

The velocity v of a particle is given in terms of time t as: $v = at + \frac{b}{t^2}$. The dimensions of a , b , and c are:

- a) $[L^2], [L], [L^{-2}]$
b) $[L^{-2}], [L], [L]$
c) $[L^{-2}], [L], [T]$
d) $[L], [L^2], [T]$

The resultant of two vectors is maximum when the angle between them is:

- a) 0°
b) 180°
c) 90°
d) 120°

charge on the sphere becomes:

- a) $-16 \mu C$
b) $16 \mu C$
c) $32 \mu C$
d) $-32 \mu C$

brought near, the force q_1 exerted on q_2 :

- a) Increases
b) Decreases
c) Remains unchanged
d) Increases if q_1 is of the same sign as q_2 and decreases if q_1 is of the opposite sign.

The force between two charges 0.06 m apart is 5 N. If each charge is moved towards the other by 0.01 m, then the force between them will be

- a) 7.20 N
b) 11.25 N
c) 22.50 N
d) 45.0 N

A charged rod attracts bits of dry cork dust, which, after touching the rod:

- a) Adhere with the rod
b) Fall down
c) Jump violently away from the rod
d) Fly upward

Answer Questions:

- a) What is meant by dimension and dimensional formula of a physical quantity? Explain with a suitable example?
b) Check the correctness of the physical equations: i) $F = 4\pi^2 f^2 m r$; F is force, f is frequency, m is mass and r is radius. ii) $F S = \frac{1}{2} m v^2 - \frac{1}{2} m v^2$; F is force, S is displacement, m is mass, u and v are initial and final velocity.

GROUP - 'A' [6x1=6]

Long Answer Questions

10. a) Explain the principle of homogeneity of dimensional analysis.
b) Convert 60 J/min into a new system of units that uses 10 cm, $\frac{1}{2}$ min, and 50 g as fundamental units of length, time, and mass by using the dimensional method.

GROUP - 'C' [2x8=16]

9. a) Why one can ignore the quantization of charge while dealing with macroscopic charge?
b) What do you mean by electrostatic induction?
c) Explain how a conductor can be charged positively by the method of induction. Is it possible to charge insulators by the method of induction?

8. a) A vector has zero magnitude. Is it meaningful to call it as a vector? Explain.
b) State triangle law of vector addition.
c) Illustrate the triangle law of vector addition graphically and analytically.

Best of Luck

11. a) What is the quantization of charge?
b) State and explain Coulomb's law of force between two stationary charges. Also write the formula for coulomb's law in vector form?
c) Two particles A and B having charges 8 pC and -2 pC respectively are held fixed with a separation of 0.20 m. Where should a third charged particle be placed so that it does not experience a net electrostatic force?
d) Two identical small conducting spheres fixed at a distance of 0.50 m. A thin conducting wire then connects the spheres. When the wire is removed, the spheres repel each other with an electrostatic force of 0.036 N. Calculate the initial charge on the spheres.

12. a) Two identical small conducting spheres fixed at a distance of 0.50 m. A thin conducting wire then connects the spheres. When the wire is removed, the spheres repel each other with an electrostatic force of 0.108 N. Calculate the initial charge on the spheres.

Time: 1:30 hrs.
 Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.
 Attempt all the questions.

MULTIPLE CHOICE QUESTIONS
 GROUP - A' [9x1=9]

- When zinc is added to CuSO_4 solution, copper is precipitated. It is so because,
 - Reduction of Zn
 - Reduction of Cu^{2+}
 - Hydrolysis of CuSO_4
 - Reduction of SO_4^{2-}
- Which of the following acts as oxidizing agent only?
 - NO_2
 - N_2O_3
 - HNO_3
 - NH_3
- Oxidation number of Hydrogen in MgH_2 is
 - +1
 - 1
 - +2
 - 2
- The increase in volume that will take place when 200 ml of gas at 300 K is heated to 360 K at constant pressure is
 - 240 ml
 - 200 ml
 - 40 ml
 - 160 ml
- A gas which perfectly obeys gas laws at all temperature is called
 - Noble gas
 - Ideal gas
 - Inert gas
 - diatomic gas
- Value of 'R' in Joule is
 - 8.314
 - 1.987
 - 0.0821
 - 8.314×10^3

- Which of the following transition involves maximum amount of energy?
- $M^-(g) \rightarrow M(g)$
 - $M(g) \rightarrow M^+(g)$
 - $M^+(g) \rightarrow M(g)^{+2}$
 - $M(g)^{+2} \rightarrow M(g)^{+3}$
- The correct order of ionization energy of C, N, O, F is
- $F > N > O > C$
 - $O > N > F > C$
 - $O > N > C > F$
 - $F > O > N > C$

SHORT ANSWER QUESTIONS
 GROUP - B' [5x4=20]

- Isotectonic structure from the following is
 - CH_3^+
 - H_2O^+
 - NH_4^+
 - CH_3^-

LONG ANSWER QUESTION
 GROUP - C' [1x8=8]

- Define electronegativity and de-electronegativity with examples. [2]
 - Find the maximum and minimum oxidation number of Bromine and Phosphorous with example. [2]
 - Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent. [2]
 - Show with examples that all reactions are not redox reactions. [2]
- $\text{KMnO}_4 + \text{HCl} \rightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{KCl} + \text{H}_2\text{O}$

Best of Luck



KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION 2082

Morning Shift Set-B

Subject: Mathematics (0071)

P.M.: 37

P.M.: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Class: XI

Time: 1.5 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [4x1=4]

Write the correct options of each question in your answer sheet.

1. What is the contrapositive of $q \Rightarrow \sim p$?

- A) $p \Rightarrow \sim q$ B) $q \Rightarrow p$ C) $q \Rightarrow \sim p$ D) $p \Rightarrow q$

2. Which one of the following is true?

- A) $A \subseteq A \cup B$ B) $A \cap B \subseteq B$ C) $A \cup \bar{A} = U$ D) All of the above

3. What is the value of $\lim_{x \rightarrow -5} x - 1$?

- A) 1 B) -1 C) undefined D) 5

4. What is the value of $\lim_{x \rightarrow \infty} \frac{\sqrt{1+25x^2}}{3+5x}$?

- A) $\frac{1}{5}$ B) 5 C) 2 D) 1

GROUP-B [5x5=25]

5. a) Determine the truth value of the statement "3 is odd and 4 is even".

[2]

Also, write its negation.

b) Construct truth table for the following statement and comment on the result $(p \Rightarrow \sim q) \wedge (p \wedge q)$

[3]

6. a) Is it possible to write $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{x} \right) = \lim_{x \rightarrow 0} \frac{1}{x} - \lim_{x \rightarrow 0} \frac{1}{x}$? Give reason.

[2]

b) Evaluate:

i) $\lim_{x \rightarrow 4} \frac{\sec y - \sec x}{\sqrt{x} - \sqrt{y}}$

ii) $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$

[2+1]

7. a) Define limit of a function $f(x)$ at $x = c$.

[1]

b) What are indeterminate forms?

[1]

c) Define union of two sets.

[1]

d) State De-Morgan's laws for logic.

[1]

e) What is the value of $\lim_{y \rightarrow 0} \frac{y}{\tan y}$?

8. a) For any two non-empty subsets P and Q of universal set U , prove that

$$\overline{P \cap Q} = \overline{P} \cup \overline{Q}$$

[2]

b) If $A \subseteq B$, then $P(A) \subseteq P(B)$. Verify this by choosing appropriate sets A and B .

[3]

9. a) For any three sets A , B , and C , prove that:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

[2+3]

b) Evaluate: $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{4x^2-\sqrt{x}+4}$

[3]

GROUP-C [8x1=8]

10. a) Does the order of elements in a set matter? Justify your answer with an appropriate example.

[2]

b) What is the difference between undefined and indeterminate forms in the context of limits? Justify your answer.

[3]

c) Give an example of a function that takes the indeterminate form $\frac{0}{0}$ but whose limit exists. Justify your answer.

[3]

Best of Luck

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
 पूर्व वैमासिक परीक्षा-२०८२

कक्षा : ११

समय : १:३० मिनेट

विभागीय सत्र 'क'

विषय : नेपाली (००११)

पूर्णांक : ३०

उत्तीर्णांक : १५

विद्यार्थीहरूले तल्लेखमा आफ्नै शब्दमा उत्तर दिनुपर्ने छ । यहाँ कित्तामा दिइएको अङ्कले पूर्णांक जनाउँछ ।

यसै प्रश्नहरू अनिवार्य छन् ।

१. दिइएका व्यञ्जन वर्णहरूको उच्चारण स्थान र प्राणत्व छुट्याई लेख्नुहोस् :

य, म, ह

२. दिइएको वाक्यलाई शुद्ध गरी पुनर्लेखन गर्नुहोस् :

बाल्यकालको दसैँको स्मृति मेरो हृदयमा अझै सम्म ताजे छ । म पूर्वी नेपालको पहाडमा जन्मेको थिएँ । मेरो गाउँ भन्दा भन्दा दसैँको यकल संस्कृतिमा जन्मेको र त्यसै रमाउँथे ।

३. दिइएको अनुच्छेदमा रेखाङ्कित पदहरूको पदवर्ग पहिचान गरी लेख्नुहोस् :

उसको सजानु हेराइ, लचकदार बिँडाइ, मिजामिलो स्वभाव सबै उत्तिकै राम्रो लाग्छ त्यसैले ऊ सरासर घर गएका बेला पनि मैले उसको बाटो हेरेको छु ।

४. दिइएका शब्दहरूलाई बहुवचनमा परिवर्तन गरी पुनर्लेखन गर्नुहोस् :

म कक्षा १२ मा पढ्ने विद्यार्थी हुँ । कलेजमा मेरो एकजना मन भिन्ने साथी छ । साथी मलाई माया मात्र होइन सहयोग पनि गर्छ । म दिनहुँ उसलाई भेट्छु ।

५. दुवै प्रश्नहरूको उत्तर दिनुहोस् :

(Y+Y= C)

क. दिइएका अनुच्छेद पढी सोधिएका प्रश्नहरूको उत्तर दिनुहोस् :

आयुर्वेद पूर्वीय प्राचीन चिकित्सा पद्धति हो । 'आयुर्वेद' शब्दको अर्थ 'आयुको ज्ञान' भन्ने हुन्छ । यसलाई आयुर्विज्ञान पनि भनिन्छ । आयुको ज्ञान भन्नाले जीवनमा पूर्ण आयु जिउने विषय र कलाको ज्ञान हो । पूर्ण आयु जिउन स्वास्थ्य पक्षी आवश्यकता हो । स्वस्थ व्यक्ति नै पूर्णायुको जीवन जिउछ भन्ने आयुर्वेदको मान्यता हो । यसैले आयुर्वेदमा जीवन स्वस्थ राख्ने उपायका बारेमा विस्तृत रूपमा चर्चा गरिएको छ । जीवन स्वस्थ राख्नु भनेको सक्तर भर शरीरमा रोगको प्रवेश हुन नदिनु हो र कर्षाचित रोगको प्रवेश भइहाले पनि उपचारका माध्यमबाट शरीरबाट त्यसको निष्कासन गर्नु हो । आयुर्वेदमा रोगको उपचारका लागि कार्याचिकित्सा, शल्य तन्त्र, शालास्य तन्त्र, भूतविद्या, कौमारभृत्य, अगद तन्त्र, रसायन तन्त्र र बाजीकरण तन्त्र गरी आठ अङ्ग रहेका छन् । बीचमध्य कार्याचिकित्सामा शारीरिक रोग, शल्य तन्त्रमा चिरफार गर्नुपर्ने रोग, शालास्य तन्त्रमा नाक, कान, घाँटी वा अन्य कुनै कारण भएको टाउकाको रोग, भूतविद्यामा भूत, गृह वा मानसिक कमजोरीका कारण लागेको मानसिक रोगको उपचारको पद्धति प्रस्तुत गरिएको छ । यस्तै कौमारभृत्यमा गर्भवती स्त्रीलाई सानै रोग र बालरोग, अगद तन्त्रमा

शरीरमा विषालु परावर्को प्रवेशका कारण उत्पन्न भएको रोग, रसायन तन्त्रमा बुढाबुढावा हुने रोग, बाजीकरण तन्त्रमा शरीरमा घातको कमीका कारण हुने रोगको उपचारको पद्धति समाविष्ट छ । आयुर्वेदका अनुसार मुकुल रूपमा शरीरमा बाल, पित्त र कफ यी त्रिदोषको असन्तुलनका कारण रोगको प्रवेश हुन्छ । यी तीन दोषबाट बच्ने मुख्य उपाय भनेको उचित आहार र उपयुक्त जीवनशैली नै हो । उपयुक्त आहार र जीवनशैलीबाट शरीरमा रोगभोग नहुने बोन वा जीवनी शक्ति बसक्त रूपमा चिकित्सीय बन्छ । जीवनी शक्तिको लक्ष्यता हुँदा शरीरमा रोगले प्रवेश नै गर्न पाउँदैन, कर्षाचित प्रवेश शरीरले पनि त्यो प्रभावहीन हुन्छ । प्रभावहीन त्यस रोगलाई पनि स्वाधीय जडीमुटी र सन्तुलको उपयोग गरी शरीरबाट सट्टै निष्कासन गर्न सकिन्छ । यसैले उचित आहार, उपयुक्त जीवनशैली र उपलब्ध स्वाधीय जडीमुटीको सेवनबाट स्वस्थ रहेर पूर्णायुको जीवन जिउन सकिन्छ भन्ने कुराको मार्गनिर्देशन गर्ने चिकित्सा पद्धति नै आयुर्वेद हो ।

प्रश्नहरू :

- आयुर्वेदका आठ अङ्ग के के हुन् ?
- कर्षाचिकित्सा र शल्य तन्त्रमा कुन कुन रोगको उपचार पद्धति प्रस्तुत गरिएको छ ?
- शरीरमा रोगको प्रवेश हुनुको कारण के हो ?
- आयुर्वेदले जीवनमा पूर्णायु जिउन के गर्नुपर्ने सुझाव दिएको छ ?

ख. दिइएको कविताका पढी सोधिएका वस्तुगत प्रश्नहरूको उत्तर लेख्नुहोस् :

असु खसाली रोइरहेथे जीवन बनका बिच
 छोडी छोडी कतै नयाँ मेरो मनका बिच
 टुटेपुटेका आशानाथि नवन दुखका गौर
 कहाँ कहाँको कल कलाको मेरो मनको पिर ।
 नालसाले योझनीको रूप गरी धारण बोली
 गहना भन्दाकै रुखी तिनी कुन कारण ।
 बिलासको यो महल मनोहर हुन्छ
 म तिमी रानी मैले बोलेँ,
 "हुन्छ सबै यो भोलि धुलो र खराती ।"
 धर्म बोलेँ, "कुन मन्दिरमा पुजा गर्छौं बाऊ"
 मैले बोलेँ, "बीरहरूको बलिदान भएको ठाउँ"
 सहानुभूतिले सङ्करण बोलिन, "तब किन रुखी भइ ?
 साथ छ को को ? घर हो तिम्रो कुन देश ?
 जान्छौ कुन पुर भवन मुसाफिर ? त्यागी कुन सन्देश ?
 "होस्ट छ मेरो गुप्त उद्योगी, साथ छ साहस बैरा
 विराच सबै घर हृदयपुरीतिर ल्याउँछु सेवा सन्देश ।"

-लक्ष्मीप्रसाद देवकोटा

प्रश्नहरू :

(क) 'म' पात्र कहाँ थिए ?

(ख) घरको अभिमानमा

(ङ) जीवनरूपी वनका विचमा

(च) 'म' पात्रले पूजा गर्ने ठाउँ कुन हो ?

(ज) वीरहरूको बलिदान भएको ठाउँ (आ) मानिसहरू भेला भएको ठाउँ

(ड) मन्दिर (ई) तीर्थ

(फ) 'म' पात्र के गर्न चाहन्थे ?

(ज) दान

(आ) धर्म

(इ) ब्रजवन

(फ) जो कविताका रचनाकार को हुन् ?

(ज) माधवप्रसाद शिर्षरे

(ङ) निर्दिष्टरूप भेट्छ

(आ) भूमि शोरवन

(ई) सत्यवीरसाद देवकोटा

(४+४=८)

दुई प्रश्नको उत्तर लेख्नुहोस् ।

क) निम्नको कवितामा पढी सोधिएका प्रश्नहरूको दिनुहोस् :

कुन पहाडले कुन बाँसाले तिमी गति छेकेछ्यो र

वीर पुर्खा ! कुन बाँडीले तिमी यात्रा रोकेछ्यो र

गर्दको भैंँ वेग तिमी कुन आकाशले बाँझ सक्थ्यो

सौरखले रच्यो नेपाल ! पहाड तराई जुट्न सक्थ्यो ।

प्रश्नहरू :

(क) नेपालीहरूको कस्तो पौरखले नेपालको रचना भएको हो ?

(ख) कविताका आधारमा हाथा वीर पुर्खाको गौरव गाथाको वर्णन गर्नुहोस् ।

ख) वनको कथामा पढी सोधिएका प्रश्नको उत्तर दिनुहोस् :

"त्यही कुरा सुनी छ, हिममात त किन हरिन्थ्यो र ! जे, एक साल भएन त बर्को साल

कहाँ जान्ना ! जब उत्तर गए रमा थे तो पिछे हटना स्यो ! तर एक कुगको दुख छ माद

! सत्यो बेलासा, किम्वद भान्न । तिमी पनि गाउँमा हुँदा हो त हामीलाई छेर बाइभरोस

मिन्थ्यो, बल मित्यो, आँखिर हामी त मर्द र गँवार न हो । तिमी सेखपड गरेको मान्छे

। तिमी भन्यो गाउँको अभिमान हो, हासो दज्जत हो । तर तिमी पनि परदेशी भट्टीदा

गाउँ मानिक्नुको एकजोटी रजाईमा पर्छे ।"

प्रश्नहरू

(क) कृष्णलाई किन सक्ने सम्मान गरेका हुन् ?

(ख) पदनेत्रका मान्छेले गाउँ छेदियो भने गाउँमा कस्तो अवस्था पर्छ ?

क) पुर्राहरूले देशका लागि के कस्तो सोपदान गरे ? समीक्षात्मक उत्तर लेख्नुहोस् । ८

सफलताको शुभकामना

Class: XI 'Science'
 Time: 1:30 hrs.
 Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.
 Attempt all the questions.

PRE-TERM EXAMINATION 2082
 DAY SHIFT 'SET-B'
 Subject: Chemistry [3011]

F.M: 37
 P.M: 15

MULTIPLE CHOICE QUESTIONS
 GROUP - 'A' [9×1=9]

1. A reducing agent is a substance which can

- a) accept electron & is oxidized
 b) donate electron & is oxidized
 c) accept electron & is reduced
 d) donate electron & is reduced

Which of the following acts as reducing agent only?

- a) SO_2
 b) H_2SO_4
 c) H_2S
 d) SO_3

Oxidation number of Pb in PbO_2 is

- a) +1, +2
 b) +2, +3
 c) +3, +4
 d) +2, +4

16 gm of O_2 at NTP would occupy a volume of

- a) 11.2 lit
 b) 22.4 lit
 c) 5.6 lit
 d) 16 lit

The density of NH_3 at 30°C and 5 atm pressure is

- a) 2.30 gm/lit
 b) 3.42 gm/lit
 c) 23 gm/lit
 d) 30 gm/lit

The curve plotted between P and V at constant temperature is called

- a) Isotherm
 b) Isobar
 c) Isochores
 d) Isomer

The size of following species increase in the order

- a) $\text{Mg}^{+2} < \text{Na}^+ < \text{F}^- < \text{Al}$
 b) $\text{F}^- < \text{Al} < \text{Na}^+ < \text{Mg}^{+2}$
 c) $\text{Al} < \text{Mg}^{+2} < \text{F}^- < \text{Na}^+$
 d) $\text{Na}^+ < \text{Al} < \text{F}^- < \text{Mg}^{+2}$

The outmost electronic configuration of the least electronegative element is

- a) $ns^2 np^5$
 b) $ns^2 np^4$
 c) $ns^2 np^3$
 d) ns^1

The correct order of ionization energy of C, N, O, F is

- a) $\text{F} > \text{O} > \text{N} > \text{C}$
 b) $\text{O} > \text{F} > \text{N} > \text{C}$

SHORT ANSWER QUESTIONS
 GROUP - 'B' [4×5=20]

1. a) Define the terms:
 i) Effective Nuclear Charge
 ii) Periodicity in Periodic Table

b) Define ionisation potential. Describe any two factors that affect ionisation potential.

2. a) Define isoelectronic species.

b) Compare size of (i) Na with Na^+ (ii) Cl with Cl^-

3. a) Derive combined gas equation.

b) Find the value of 'R' in SI unit.

c) What is the significance of 'R'?

4. a) Write any two applications of Boyle's law.

b) A spherical balloon of 21 cm diameter is to be filled with hydrogen gas at NTP from a cylinder containing gas at 20 atm and 27°C . If the cylinder can hold 2.82 litre of water vapour at NTP, calculate the number of balloons that can be filled up.

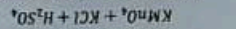
LONG ANSWER QUESTION
 GROUP - 'C' [1×8=8]

1. a) Differentiate between oxidation number and valency.

b) Define auto-redox reaction with example.

c) Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent.

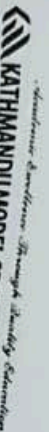
d) Find the oxidation number of



- i) S in H_2SO_4
 ii) Cr in CrO_3

"Best of Luck"

c) $\text{O} > \text{N} > \text{F} > \text{C}$
 d) $\text{F} > \text{N} > \text{O} > \text{C}$



KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION- 2082

Class: XI
Date: _____

Subject: Computer Science [4271]

Students are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

F. M.: 25
P. M.: 10

Multiple Choice Questions.

GROUP - 'A' [5x1=5]

What is smallest unit of the information?

- a) A bit b) A byte c) A block d) A nibble

Which generation of computers saw the development of minicomputers and the emergence of time-sharing operating systems?

- a) First b) Second c) Fourth d) Third

Which of the following is typically used to store the computer's boot-up instructions and firmware?

- a) Read-Only Memory (ROM) b) Random Access Memory (RAM)
c) Hard Disk Drive (HDD) d) Solid State Drive (SSD)

A CPU with a clock speed of 3.5 GHz can perform how many clock cycles per second?

- a) 3.5 million cycles b) 3.5 billion cycles
c) 3.5 trillion cycles d) 350 million cycles

A device that receives data from the computer and displays it in a human-readable form is called a(n) _____ device.?

- a) Processing b) Storage c) Input d) Output

GROUP - 'B' [3x4=12]

Short Answer Questions

What is ROM? Explain different types of ROM in details.

[1+3]

How do you define term Hardware, Software and Firmware? Give one example of each.

[3+1]

OR

Differentiate between Analog and Digital Computer.

[4]

3. Write down the steps to create folder and file in Windows Operating system.

[4]

Long Answer Questions.

GROUP - 'C' [1x8=8]

1. What do you mean by generation of computer? Describe different generations with all the main features included in each.

[2+6]

"Best of Luck"

- d) What warnings does corona give to human beings? Explain. (Corona Says)
- Write a readable story using the clues given below:
- cats ... find a piece of bread ... quarrel over it ... monkey offers to divide ... cats a little from each side to make equal ... finishes bread ... cats fooled ... moral
- Rewrite the sentences as indicated in the brackets.
- [3x1=3]
- a) I have bought the car. (Put the adverb *already* in the correct place.)
 b) She looks pretty. (Put the adverb *fairly* in the correct place.)
 c) They visit America. (Put the adverb *now and again* in the correct place.)

Choose the best options:

- [5x1=5]
- a) The prefix 'un-' goes with the word ...
 i) happy ii) legal iii) possible iv) correct
- b) The meaning of the prefix in the word 'misuse' is
 i) before ii) again iii) wrong iv) not
- c) The dress is very **elegant**. Choose the correct meaning of the word 'elegant'.
 i) clumsy ii) beautiful iii) heavy iv) dirty
- d) If the guide words are "**crop - crush**", which of the following will appear on that page?
 i) cry ii) cross iii) crust iv) crow
- e) 'Stone' comes before 'Story' because comes before in the alphabet. (Fill in the blanks with suitable letters)
 i) n/r ii) r/n iii) e/y iv) y/e

[Best Wishes]

Page: XI
 Candidates are required to give their answers in their own words as far as practicable. The figures in margins indicate full marks.

Multiple Choice Questions.

GROUP - 'A' [5x1=5]

- How many nibble(s) equals to 1 byte?
 a) 4
 b) 1
 c) 2
 d) 1024
- The introduction of the graphical user interface (GUI) and the mouse is a key feature of which generation?
 a) Fifth
 b) Second
 c) Fourth
 d) Third
- Short Cut key to rename the file's name is:
 a) F1
 b) F2
 c) F3
 d) none of these
- Why is the 'word size' (e.g., 32-bit vs. 64-bit) a factor in a computer's speed?
 a) It determines the physical size of the CPU chip
 b) It dictates the type of monitor that can be connected
 c) It controls the speed of the internet connection
 d) A larger word size allows the CPU to process more data in a single operation
- Which of these devices can be used for both input and output?
 a) Webcam
 b) Scanner
 c) Plotter
 d) Touchscreen

Answer Questions.

GROUP - 'B' [3x4=12]

- [1+3] What is Computer Bus system? Explain its types.
 [2+2] What is Explain the difference between a margin and a page break. Where would you find the options to adjust these in Word?
 'OR'
 [4] Explain the features and benefits of using MS word for document creating and editing.

Long Answer Questions.

GROUP - 'C' [1x8=8]

3. What is Mobile Computing? Explain its advantages and usability. [2+2]
1. Explain the evolution of computer, describing the technologies used in different generations of computer.
 [8]
- "Best of Luck"

Students are required to give their answer in their own words as far as practicable.
 Figure in the margin indicates full marks:
 Time: 2hrs.

Attempt all questions.

Read the following excerpt and do the activities that follow.

Alyohin often thought of Anna Alexeyevna, the young, intelligent, and graceful wife of Luganovitch. They had met at a dinner in the town, shortly after a legal case had exhausted Alyohin. She was striking, with cordial, intelligent eyes that made him feel immediately close and familiar. Her charm and warmth impressed him deeply, and though she was married, a gentle connection began to grow between them.

During the summer, Alyohin spent long hours at Sofino, the estate where he worked. Though busy with farming and managing the estate, his thoughts often drifted to Anna Alexeyevna. When he visited the town, he always went to see her. She would greet him anxiously, asking why he had been absent, and her eyes, her voice, her step, always made him feel something extraordinary. They talked for hours, sometimes in silence, sometimes at the piano, sometimes playing with her child, yet never confessing the love that grew between them.

Alyohin struggled with his feelings, thinking of her husband, her children, and the life they shared. He knew that to express his love would disrupt their household and bring dishonor. Anna Alexeyevna seemed to reason the same way. They both silently reassured the moments together, yet their affection remained hidden, guarded by propriety and caution.

Years passed. Anna Alexeyevna became more withdrawn, suffering from low spirits and neurasthenia, while Alyohin continued to visit. They sat side by side at the theatre, shoulders touching, feeling the closeness of hearts unspoken. At last, the day of parting came. At the train station, Alyohin ran to her compartment to return a forgotten basket. Their eyes met, and all restraint vanished. She pressed her face to his chest, tears flowed freely, and he confessed his love. The train moved on. He kissed

A. Answer the following questions in a single sentence each: [5X3=3]
 a) Who was Anna Alexeyevna?
 b) How did Alyohin feel when he first saw Anna Alexeyevna?
 c) Why did not Alyohin and Anna Alexeyevna openly confess their love?
 d) How did Alyohin and Anna Alexeyevna spend time together?
 e) How did Alyohin express his feeling for Anna at the train station?

B. Fill in the blanks with suitable information from the passage: [5X1=5]
 a) Alyohin first met Anna Alexeyevna at a _____ hosted by her husband.
 b) Alyohin often spent long hours at _____, thinking of Anna Alexeyevna.
 c) At the theatre, they used to sit side by side, with their _____ touching.
 d) Anna suffered from low spirits and was being treated for _____ in her later years.
 e) At the train station, Anna had almost forgotten a _____ [5]

2. Explain with reference to the context: [5]
 a) You call yourself 'the crown of creation';
 What about the others.
 Those who fly in the sky, Those who live in the ocean,
 Those who crawl on the earth,
 And those, the sources of your oxygen?
 You think they are your slaves
 Who you can sell or kill
 At your will.

3. Answer the following questions in about 75 words each. [3X2=6]
 a) What are symbols? Describe the symbols used in the story, *The Selfish Giant*.
 b) Describe the Giant's garden. (*The Selfish Giant*)

It is impossible to say how first the idea entered my head, but once it did, I knew I could not rest until it was carried out. You see, there was an old man in the house where I lived. He was kind and gentle, and had never wronged me in any way. Yet his eye—oh, his eye!—it was like the eye of a vulture, pale blue, with a film over it. Whenever he looked at me, my blood ran cold. I decided that I must rid myself of that eye forever. Every night, I crept into his room as he slept, careful not to awaken him. I moved silently, like a shadow, listening to his even breathing. For seven nights, I could not kill him as his eyes were closed. At first, the task seemed impossible, but I waited patiently. Finally, on the eighth night, I saw my chance. The old man's eye was open, staring at me with that ghastly glare. My heart pounded wildly, yet I moved with the utmost care. Suddenly, a faint sound made me pause—it was the sound of his heartbeat, growing louder and louder in my ears. The fear, the suspense, was unbearable. In a frenzy of horror and obsession, I killed him. I struck swiftly, silencing the eye forever. The old man lay still. At first, I felt relief, then a strange unease took hold. I hid the body carefully under the floorboards, confident that no one would ever discover my deed. But soon, the sound began again. A dull, steady thumping filled my mind. At first, I told myself it was my imagination, but the sound grew louder, echoing in my skull. It was the beating of the old man's heart! No one could hear it, yet it tortured me. In the end, I could bear it no longer. Driven mad by guilt, I confessed everything to the authorities.

- A. Answer the following questions in a single sentence each. [5x1=5]**
- Why did the narrator decide to kill the old man?

- Why was the narrator unsuccessful in killing the old man for seven nights?
- Where did the narrator hide the old man's body?
- Why did the narrator confess to the authorities?
- Write a suitable title for the passage.

B. Fill in the blanks with suitable information from the passage. [5x1=5]

- The old man had never _____ the narrator in any way.
- The eye of the old man was pale blue, like the eye of a _____.
- The narrator crept into the old man's room every night while he was _____.
- The narrator moved like a _____ when he entered the old man's room.
- The narrator heard a _____ sound that filled his mind.

2. Explain with reference to the context: [5]

- I will depart one day.
But remember
There're many others like me.
They'll come too.
If you don't get rid of your inflated ego,
You'll be back to your cave time
That you endured
Long, long, long ago —

3. Answer the following questions in about 75 words each. [3x2=6]

- What is the theme of the story, *The Selfish Giant*? Describe.
- Who is the little child compared to? Discuss. (*The Selfish Giant*)
- What message does Corona give to humankind? (*Corona Says*)

4. Write a readable story using the clues given below: [2]

two friends walking through forest — bear comes — one climbs tree — other lies down — bear sniffs and goes away — friend on tree asks what bear whispered — "Never trust a selfish friend." — moral

rewrite the sentences as indicated in the brackets.

- 2) She has finished her homework. (Put the adverb *just* in the correct place.) [3x1=3]

- 3) The boy is intelligent. (Put the adverb *fairly* in the correct place.)
4) We go for a walk. (Put the adverb *rarely* in the correct place.)

Choose the best options:

- 5) The prefix 'im-' goes with the word ...
i) patient ii) legal iii) correct iv) happy
6) The meaning of the prefix in the word '**repaid**' is ...
i) wrong ii) again iii) before iv) not
7) The food was very **delicious**. Choose the correct meaning of the word 'delicious'.
i) tasteless ii) mouthwatering iii) dirty iv) spicy
8) If the guide words are '**moon - more**', which of the following will appear on that page?
i) mouse ii) morning iii) move iv) moral
9) 'Plant comes before "Plant" because ... comes before ... in the alphabet. (Fill in the blanks with suitable letters)
i) n/t ii) e/n iii) a/e iv) e/t

[Best Wishes]

15 hrs.
 Answers are required to give their answers in their own words as far as practicable.
 Report all the questions.

GROUP - A [MCQ] (4x1=4)

1. The correct options of each question in your answer sheet.
 Which one of the following is logically equivalent to $\sim (p \vee \sim q)$?
 (A) $p \wedge q$ (B) $\sim p \wedge \sim q$ (C) $\sim p \vee \sim q$ (D) $p \wedge q$

2. What is the value of $\lim_{x \rightarrow \infty} \frac{x^2 + 16x + 4}{x}$?
 (A) 4 (B) 2 (C) 4 (D) 8

3. Which one of the following is true?
 (A) $A \subseteq A \cup B$ (B) $A \cap B \subseteq A$ (C) $A \cap A = \phi$ (D) all of the above

4. What is the value of $\lim_{x \rightarrow 0} \frac{x}{x+y}$?
 (A) 0 (B) 0 (C) undefined (D) y

GROUP - B (5x5=25)

1. Define intersections of two sets with an example.
 Define contradiction of a statement.
 What is the value of $\lim_{x \rightarrow 0} \frac{x^2 + \ln(1+x)}{x}$?
 What the value of $\lim_{x \rightarrow 0} \frac{x}{x-1}$?
 Write the negation of $p \Rightarrow q$.

[1+1+1+1+1=5]

2. Evaluate:
 a) $\lim_{x \rightarrow 0} \frac{x \tan(x+y) - y \tan x}{x}$
 b) $\lim_{x \rightarrow y} \frac{\frac{x-y}{\sqrt{x}-\sqrt{y}}}{\frac{x-y}{\sqrt{x}-\sqrt{y}}}$

[2+3]

3. Define set. How does it differ from collection? Justify your answer. [2]
 For any two non-empty subsets M and N of universal set U , prove that:
 i) $M \cup N = M \cup N$ ii) $M \cap N = M \cap N$
 [1.5+1.5=3]

Best of Luck

GROUP - C (8x1=8)

11. a) Is it possible to write
 $\lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} [f(x) \times g(x)]$?
 Explain with an example.

b) Under what condition are two statements logically equivalent? Justify your answer. [3]
 c) Does the order of objects in a set matter? Explain. [2]

8. Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 4}{x - 2}$. [2+2+1=5]
 9. a) Give an example of a function which takes the indeterminate form $\frac{0}{0}$.
 b) Using algebraic formula.
 c) Is it possible to find $f(3)$? Explain.
 10. a) Define power set with an example.
 b) Evaluate $\lim_{x \rightarrow 0} \sqrt{x}(\sqrt{x} - \sqrt{x-4})$.
 c) Construct truth table for the statement $(p \Rightarrow q) \vee (p \wedge \sim q)$ and comment on the result. [3]

- (क) एक अक्षरोंपर कुराकरी गई रमाइन
(ख) सगरीसो मिलेर बौछी याका पूरा फरे
(ग) अन्त्याई सारसोम गने
(घ) दो कविताका रचनाकार को हुन ?
(ङ) लक्ष्मीप्रसाद देवकोटा
(च) माधवप्रसाद मिश्र

(क) विन्ध्य दक्षिण
(ख) गोपालप्रसाद मिश्र

हुने प्रश्नको उत्तर लेबुहोस् ।

क) हिंदुएको कवितामा पछि संक्षिप्तका प्रारम्भको विबुहोस् ।

(१५४८८)

कुन पछाडले कुन बाँसाले तिम्रो नाँत छेकेको र
कीर पुर्खा । कुन आँधीले तिम्रो यात्रा रोकेको र
गण्डको भौं बेग तिम्रो कुन आकाशले बाँध्न सको
पीरखले रत्यो नेपाल । पछाड तराई जुट्न सको ।

प्रश्नहरू :

- (अ) नेपालीहरूको कस्तो पीरखले नेपालको रचना भएको हो ?
(आ) कवितामाका आधारमा हावा कीर पुर्खाको गौरव गाथाको अर्थन गर्नुहोस् ।

ख) तलको कथामा पछि संक्षिप्तका प्रश्नको उत्तर विबुहोस् :

"त्यही कुरा चुली छ, हिमाल त किन हाँस्यो र । अरे, एक काल भएन त बर्को बास
कहाँ जाँदा । जब उत्तर गए रमा थुँगे भिँडे हाँसा बसो ? तर एक कुराको टुक्छ बाँद
। यस्तो बेलामा, किनून भाँद । तिम्रो पनि गाउँमा हुँदा हो त हाँसीलाई धेरै बाढभोग्न
मिल्थ्यो, घन भिँड्यो, आँधिर हाँसी त मुहुरे र नौबार न हो । तिम्रो नेचुरम भर्को गाँह
। तिम्रो भन्या गाउँको अभिमान हो, हाँसा हुँजस्त हो । तर तिम्रो पनि पदरेनी पछिँदा
गाउँ माथिँकलको एकलौटी रजाईमा पर्छ..."

प्रश्नहरू

- (अ) कथामाई किन बढीने सम्मान गरेका हुन ?
(आ) पदरेनीकोका मान्छेले गाउँ छोड्यो भने गाउँमा यस्तो अलर पर्छ ?

गीत पुर्खाहरूले गरेका सोपानको बारेमा कसोलाएक उत्तर लेबुहोस् ।



KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION 2082

Gr. XI

at 1.5 hrs.

Subject: Mathematics (0071)

Students are required to give their answers in their own words as far as possible. The figures in the margin indicate full marks.

Day Shift SET-A

F.M.: 37

P.M.: 15

attempt all the questions.

GROUP - A [MCQ] [4x1=4]

Write the correct options of each question in your answer sheet.

Which of the following is logically equivalent to $\sim (p \wedge \sim q)$?

- A) $\sim p \wedge q$ B) $\sim p \vee q$ C) $\sim p \vee \sim q$ D) $p \vee \sim q$

If $A = \{a, b, c, d\}$, then the total number of nonempty proper subsets is

- A) 15 B) 14 C) 16 D) 13

What is the value of $\lim_{x \rightarrow 0} \frac{x^2}{x^2 - x}$?

- A) 0 B) $\frac{1}{0}$ C) ∞ D) $-\infty$

What is the value of $\lim_{x \rightarrow 2} \sqrt{x}$?

- A) $\sqrt{2}$ B) 2 C) undefined D) $1/\sqrt{2}$

GROUP - B [5x5=25]

1. Define difference of two sets.

2. State De-morgan laws for logic.

3. What is the value of $\lim_{x \rightarrow 0} \frac{\tan(x-\theta)}{x+\theta}$?

4. What is the value of $\lim_{x \rightarrow 0} \frac{2x-1}{x}$?

5. Define conditional statement.

[1+1+1+1+5]

1. Evaluate: a) $\lim_{x \rightarrow 0} \frac{\sqrt{2x+1}-\sqrt{x+1}}{\sqrt{x}}$

b) $\lim_{y \rightarrow 0} \frac{\sqrt{\sin(x+y)}-\sqrt{\sin x}}{y}$

[2+3]

2. Define set. How does it differ from collection? Justify your answer. [2]

3. For any two non-empty subsets X and Y of universal set U , prove that:

$$i) \overline{X \cap Y} = \overline{X} \cup \overline{Y}$$

$$ii) \overline{X \cup Y} = \overline{X} \cap \overline{Y}$$

[1.5+1.5=3]

10. a) If $f(x) = \frac{\sin(x-a)}{x^2-a^2}$, then

i) Determine $f'(a)$, if it exists. [1]

ii) Examine whether $\lim_{x \rightarrow a} f(x)$ exist? Justify your answer. [1]

b) Evaluate: $\lim_{x \rightarrow \sqrt{5}} \frac{x^2-5}{x^2+5\sqrt{5}x-30}$ [3]

11. a) Define power set with an example. [2]

b) Construct truth table for the statement $(p \vee q) \vee (\sim p)$ and comment on the result. [3]

GROUP - C [8x1=8]

12. a) Is it possible to write

$$\lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} [f(x) + g(x)]?$$

Explain with an example. [3]

b) Under what condition are two statements logically equivalent? Justify your answer. [3]

c) Does the repetition of elements in a set affect the set? Explain. [2]

Best of Luck

(A) चरक के भाष्य की शुरुआत का प्रमाण उपलब्ध है :

(४) माधुके कविश्रीमा राम पर्वतक बामु क मायुका
(५) राम पर्वतक मायु क कविश्रीमा राम पर्वतक

[illegible]

१३. धर्मनको उत्तर लेख्नुहोस् ।
 (क) दिव्यको कवितामा पढी समिएका धर्मनहरूको दिव्यताः

[illegible][illegible]

ԱՐԽԻՎԻ ԽՈՐՀԱՐ

७. ही प्रणाली का सी पीएम द्वारा प्रकृत प्रमाणित रूप में।

1. The above information is correct. (X)
2. The above information is not correct. (X)
3. The above information is not correct. (X)
4. The above information is not correct. (X)

[illegible]

पूर्व वैमासिक परीक्षा-२०८२

क्षा : ११

अव : १:३० मिनेट

विषय : नेपाली

(००११)

पुर्णाङ्क : ३७

उत्तीर्णाङ्क : १५

प्रश्नहरू सकेसम्म आफ्नै शब्दमा उत्तर दिनुपर्ने छ । दायाँ किनारामा दिइएको अङ्कले पुर्णाङ्क बताउँदछ ।

१. प्रश्नहरू अनिवार्य छन् ।

दिइएका व्यञ्जन वर्णहरूको उच्चारण स्थान र प्राणत्व छुट्याई लेख्नुहोस् :

ख, त, प

दिइएको वाक्यलाई शुद्ध गरी पुनर्लेखन गर्नुहोस् :

३३ वाक्यकालको दसैको स्मृति मेरो हृदयमा अझ सम्म ताजै छ । म पुर्याँ नेपालको पहाडमा जन्मेकि भिएँ । मेरो गाउँ भन्डै भन्डै दसैको यकल सस्कृतिमा जम्प्यो र त्यसमै रमाउँथ्यो ।

दिइएको वनुच्छेदमा रेखाङ्कित पदहरूको पदवर्ग पहिचान गरी लेख्नुहोस् :

मानिसको शोभा र सम्पत्ताको रक्षा गर्ने कुरा साज हो । त्यो साज मानिसमा छैन भने त्यस नाइगो मानिसित कसको के लाग्छ र ?

दिइएका वाक्यहरूलाई बहुवचनमा परिवर्तन गरी पुनर्लेखन गर्नुहोस् :

मैले गाउँमा वास्यकालको साधीलाई भेटे । मैले जीवन र जगतका बारेमा धेरै बेर कुरा गरे । साधीले आफ्नो प्रगतिको कथा सुनायो । म जित्छु परेँ ।

दुवै प्रश्नहरूको उत्तर दिनुहोस् :

(४+४=८)

क. दिइएका वनुच्छेद पढी सोधिएका प्रश्नहरूको उत्तर दिनुहोस् :

मानिस सासारिक वस्तुहरूको अनुभव पाँच ज्ञानेन्द्रियबाट गर्दछ । उसका मुख्य नाडीहरू मस्तिष्कसँग मिल्नेका हुन्छन् । आँखा, नाक, कान, जिह्वो र छाला ज्ञानेन्द्रियहरू हुन् । मानिसले ससारको अनुभव धिने पाँच इन्द्रियहरूले गर्दछन् । यी इन्द्रियहरूलाई हामीले सधैं स्वच्छ र सरल राख्नु पर्दछ । सामर्थ्यभन्दा अधिक काम तिनाले इन्द्रियहरूमाथि आघात पुग्दछ । फलस्वरूप मस्तिष्कको क्षमता कमजोर हुन गई मानिसको स्मरणशक्ति कम हुन जान्छ । मस्तिष्कलाई सबल र स्वस्थ बनाउन शुद्ध रगतको आवश्यकता पर्दछ । ठूला ठूला महात्मा एवम् बुद्धिमान् पुरुष वीर्यको रक्षा गर्दा गर्दै पनि निर्वल प्रतीत हुन्छन् । यसको कारण के हो भने तिनले मस्तिष्कबाट धेरै काम लिनु पर्दछ, जसबाट रक्तादि पोषक भाग विचार शक्तिको साथ अधिक खर्च हुन्छ । रात दिनको पढाइको

कामले विद्यार्थीको शारीरिक बल पाँच लपट घाय हुन जान्छ । मस्तिष्कलाई बलवान् बनाउनको लागि सत्वगुणी खाद्य पदार्थको सेवन गर्नुपर्छ । दुध, घिउ, चामल, गहुँ, केरा, अदूर, स्याउ आदि पदार्थमा सत्वगुण रहेको हुन्छ । तमोगुणी पदार्थले शरीरलाई मोटो बनाउँछ भने रजोगुणी पदार्थले शक्तिलाई शीघ्र व्यय गरिदिन्छ । त्यसैले यस्ता पदार्थको प्रयोग कम गर्दै जानुपर्छ ।

क) मानिसले ससारको अनुभव केबाट गर्छ ?

ख) मानिसको मस्तिष्कलाई सबल राख्ने उपायहरू के-के हुन् ?

ग) मस्तिष्कको क्षमता कमजोर हुनाको कारणहरू के-के हुन् ?

घ) शीघ्र र सामर्थ्य शब्दको अर्थ लेख्नुहोस् ।

ख) दिइएको कवितामा पढी सोधिएका वस्तुगत प्रश्नहरूको उत्तर लेख्नुहोस् :

यहाँ पनि

म जस्तै अर्को नदी रहेछ ।

दोभानमा भेट भएपछि

दुई फरक बाटाबाट आएका नदीहरू

आफू जस्तै अर्को नदी देखेर छक्क परे ।

दुवैको यात्रा उस्तै थियो

सङ्घर्षले भरिएको बाटो

उस्तै थियो हृदयको सङ्गीत उस्तै थियो

प्रेमको सपना उस्तै थियो ।

कसलाई जित्नु छ र अब ?

दोभानमा भेट भएपछि

बाँकी यात्रा पूरा गर्न

उनीहरू सँगसँगै मिलेर बग्न थाले । - विष्णु वडाल

प्रश्नहरू :

(क) नदीहरू किन छक्क परे ?

(अ) आफू जस्तै खोला देखेर

(आ) विचित्र स्वरूपका मान्छे देखेर

(इ) एउटा माछाले अर्को माछालाई निलेको देखेर

(ई) आफू जस्तै अर्को नदी देखेर

(ख) दुई नदीका बिच के के कुरामा समानता छ ?

(अ) सङ्घर्ष, सङ्गीत र व्यथा

(आ) सङ्घर्ष, सङ्गीत र सपना

(इ) सुख, समृद्धि र सपना

(ई) सुख, समृद्धि र स्वभाव

(ग) दुई नदी किन सँगसँगै बग्न थाले ?

(अ) सुखदुःखका भावना साटासाट गर्न



KATHMANDU MODEL SECONDARY SCHOOL

PRE - TERM EXAMINATION 2082

Sr. XI 'Science'
F.M: 37
P.M: 15
DAY SHIFT 'SET-B'
Subject: Physics [1011]

Students are required to give their answers in their own words as far as practicable. The figures in margin indicate full marks.

Attempts all the questions.

GROUP - 'A' [6x1=6]

Multiple Choice Questions:

The equation of a wave is given by, $y = A \sin \left[\omega \left(\frac{z}{v} - k \right) \right]$, when ω is

angular velocity, x is position and v is linear velocity. The dimensional formula for k is:

- a) $[L T^{-1}]$
- b) $[T]$
- c) $[T^{-1}]$
- d) $[T^2]$

Identify the correct statement:

- a) Velocity, displacement, acceleration, force are vectors
- b) Weight, mass, torque and energy are scalars
- c) Angular velocity, momentum and electric current are scalars
- d) A vector quantity is one which has magnitude only whereas scalar quantity is one which has both magnitude and direction.

If 2×10^{13} electrons are added to a neutral metal sphere, the charge on the sphere becomes:

- a) $-1.6 \mu C$
- b) $1.6 \mu C$
- c) $3.2 \mu C$
- d) $-3.2 \mu C$

A soap bubble is given a negative charge, then its radius:

- a) Decreases
- b) Increases
- c) Remains unchanged
- d) Nothing can be predicted as information is insufficient.

The force between two charges 0.06 m apart is 5 N . If each charge is moved towards the other by 0.01 m , then the force between them will be

- a) 7.20 N
- b) 11.25 N
- c) 22.50 N
- d) 45.0 N

When a positively charged conductor is connected to earth:

- a) No charge flow occurs
- b) Electrons flow from conductor to earth
- c) Protons flow from conductor to earth
- d) Electrons flow from earth to conductor

GROUP - 'B' [5x3=15]

Answer Questions:

What are the basic rules of dimensional analysis?

[1]

b) Can we tell the unit of a physical quantity from its dimensions? What about dimensions from the unit of a physical quantity? [1]

c) Check the correctness of the physical equations: (i) $t = 2\pi \sqrt{\frac{l}{g}}$; t is time period, l is length and g is acceleration due to gravity (ii) $T^2 = \frac{4\pi^2 a^3}{GM}$; T is time period, G is gravitational constant, M is mass and a is radius of the orbit. [3]

g. a) What is meant by resultant vector. [1]

b) Two vectors acting on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangular representation? State and explain the law involved. [3]

c) Under what condition is the magnitude of resultant vector is zero vector? [1]

9. a) Explain about Action of point with a demonstration? [2]

b) What do you mean by electrostatic induction? How will you give permanent positive charge to a conductor by the method of induction? Explain in detail. [3]

GROUP - 'C' [2x8=16]

Long Answer Questions:

10. a) What do you mean by dimension of a physical quantity? What is the difference between dimensions and unit of a physical quantity? [2]

b) A body moving through air at a speed v experiences a retarding force F given by the equation $F = kA\rho v^2$, where A is the surface area of the body, ρ is the density of air and k is dimensionless constant. Deduce the value of x . [2]

c) Convert 60 J/s into a new system of unit which has 10 gm , 10 cm and 10 second as fundamental units of mass, length and time. [2]

d) Is it possible to derive all the physical equations by the method of dimensional method? Explain. [2]

11. a) Write the formula for Coulomb's force in vector form? What is the significance while expressing in vector form? [2]

b) What is electrical permittivity? Write its value and unit for vacuum in SI and cgs system of unit. [2]

c) If a body gives out 10^5 electrons per second, how much time is required to get a total charge of 1 C from it? What conclusion would you draw from the result obtained? [2]

d) Two positive point charges are at a separation of 0.3 m repel each other with a force of 1.6 N . If the sum of two charges be 10 microcoulomb , what are their separate values? [2]

Best of Luck

Students are required to give their answers in their own words as far as practicable. The figures in margin indicate full marks.

Attempt all the questions.

Multiple Choice Questions:

- The correct formula for the displacement of a particle executing periodic motion is (v, T, a, y and t represent velocity, time period, amplitude, displacement and time respectively)
- $y = a \sin \frac{2\pi}{T} t$
 - $y = a \sin \frac{2\pi}{T} t$
 - $y = a \sin \frac{2\pi}{T} t$
 - $y = a \sin \frac{2\pi}{T} t$

GROUP - 'A' [6x1=6]

- A vector remains unchanged on:
- Rotating it through some angle
 - Taking its dot product with a unit vector
 - Taking its cross product with a unit vector
 - Shifting it parallel to itself

- When 2×10^{14} electrons are removed from a neutral metal sphere, the charge on the sphere becomes:
- 16 μC
 - 16 μC
 - 32 μC
 - 32 μC

- A charge q_1 exerts some force on a second charge q_2 . If a third charge q_3 is brought near, the force q_1 exerts on q_2 :
- Decreases
 - Increases
 - Remains unchanged
 - Increases if q_3 is of same sign as q_1 and decreases if q_3 is of the opposite sign.

- The dielectric constant for a metal is:
- Greater than 1
 - Infinity
 - 0
 - 1

- You are travelling in a car during a thunderstorm. In order to protect yourself from lightning would you prefer to:
- Take shelter under a tree
 - Touch the nearest electric pole
 - Remain in the car
 - Get out and lie flat on the ground

GROUP - 'B' [5x3=15]

Answer Questions:
 Do all the physical quantities have dimensions? Justify your answer.
 In CGS system, the value of Stefan's constant σ is $5.67 \times 10^{-8} \text{ erg s}^{-2} \text{ cm}^{-2} \text{ K}^{-4}$. Find its value in international system of unit.

Long Answer Questions

- a) A physical equation is dimensionally correct. Is it necessary that the equation is physically correct?
 b) Taking force (F), length (L), time (T), and current (A) as fundamental quantities, find the dimensional formula for permeability μ_0 .
 c) The gravitational constant is $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$. Express G in terms of length in astronomical units (1 AU = $1.496 \times 10^{11} \text{ m}$). Mass in solar masses (1 $M_\odot = 1.989 \times 10^{30} \text{ kg}$). Time in years by dimensional method.
 d) While deriving the relationship between physical quantities by dimensional analysis, dimensional constants enter into a relationship. Can you find its magnitude by the method of dimensions? If not, how will you find?
- a) Mention any two limitations of the dimensional method.
 b) State and explain Coulomb's law of force between two stationary charges. Also, write the formula for Coulomb's force in vector form.
 c) What is the significance while expressing in vector form?
 d) Two particles A and B having charges 4 nC and -1 nC respectively are held fixed with a separation of 0.20 m. Where should a third charged particle be placed so that it does not experience a net electrostatic force.
 e) Two identical spheres having unequal opposite charges are placed at a distance 30 cm apart. After touching them mutually, they are again separated at the same distance apart. Now they repel each other with a force of 0.025 N. Calculate the final charge on each of them.

GROUP - 'C' [2x8=16]

- a) Explain in detail.
 b) permanent negative charge to a conductor by the method of induction?
 c) What do you mean by electrostatic induction? How will you give permanent negative charge to a conductor by the method of induction?
 d) Repulsion is the surest test of electrification. Explain.
 e) Under what condition is the magnitude of resultant force is zero Newton?
- a) Under what condition is the magnitude of resultant force is zero Newton?
 b) Under what condition is the magnitude of resultant force is zero Newton?
 c) Under what condition is the magnitude of resultant force is zero Newton?
 d) Under what condition is the magnitude of resultant force is zero Newton?
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 e) Under what condition is the magnitude of resultant force is zero Newton?

Best of Luck